

Returns to Scale, Productivity, and Markup: Revisit the Export Premium*

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First Draft: August 23, 2022

This Draft: July 31, 2026

Abstract

Many policies and firm investments directly influence firms' market power and production costs beyond productivity. This paper develops a new method for estimating firm-level markups, productivity, and returns to scale using revenue data and applies it to examine the gains from exports. We show that increasing returns to scale and markups are key sources of export gains, alongside productivity. Exports generate significant efficiency gains in Chinese manufacturing, with nearly two-thirds attributable to increasing returns to scale and the remainder to productivity gains. Exporters also charge higher markups but offer lower prices, raising profits by about 25% and benefiting consumers. Our findings also explain why traditional revenue-based productivity may miss the export premium even when exports increase markups and efficiency.

Keywords: Export premium, returns to scale, productivity, markup, production function estimation

*The authors thank Jo Van Biesebroeck, Jan De Loecker, Jiandong Ju, Bingjing Li, Shengyu Li, Yao Amber Li, Jiawei Mo, Larry Dongxiao Qiu, Mark Roberts, Chang Sun, Heiwai Tang, James R. Tybout, Frank Verboven, Frederic Warzynski, Miaojie Yu, Xiaobo Zhang, Yifan Zhang, and Xiaodong Zhu for their insightful comments. The authors also benefited from discussions with participants in the 2024 Econometric Society European Meeting, the 2024 Asia Meeting of the Econometric Society, 2024 North American Summer Meeting of the Econometric Society, 2024 China Industrial Organization conference, 2024 International Association for Applied Econometrics, 2023 Hong Kong Economic Association Biennial Conference, 2023 HKUST Conference on International Economics, 2023 Danish International Economics Workshop, and the 13th Annual Meeting of China Trade Research Group. This research was partly work was supported by the General Research Fund of Hong Kong (No.17501521) and the National Natural Science Foundation of China (No.72503246).

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1 Introduction

Many policies and firm investments directly affect firms’ market power and production efficiency beyond their effects on productivity. Yet empirical analyses in the literature overwhelmingly emphasize effects on productivity, rather than on market power and other efficiency channels, such as returns to scale. This focus can introduce bias into evaluations of the impact of policy changes and firm investments. Take exporting as an example. Exporting may enhance a firm’s productivity and markup¹. If production exhibits increasing returns to scale, the output expansion associated with exporting further lowers marginal costs beyond the direct productivity effect. However, empirical studies based on revenue productivity (TFPR) have often found mixed or limited effects of exporting (see [Bernard and Jensen, 1999](#); [Keller, 2004](#), for a review), potentially due to biases arising from ignoring the roles of markups and returns to scale. This paper investigates the multiple sources of export premium—namely, markup, increasing returns to scale, and productivity—using a large dataset of Chinese manufacturing firms.

This study is motivated by two facts observed in a large dataset from the Chinese manufacturing sector. First, exports substantially increase firms’ markup, as proxied by the revenue-to-variable costs ratio. Second, export reduces prices substantially, as shown in a smaller dataset with output prices, consistent with [De Loecker et al. \(2016a\)](#) and [Garcia-Marin and Voigtländer \(2019\)](#). The increase in markup and decline in prices directly imply a gain in production efficiency from exports. However, we find a negligible effect of export on traditional revenue-based productivity. [Garcia-Marin and Voigtländer \(2019\)](#) noticed the role of markup when evaluating the effect of export on productivity. Under constant returns to scale, they show that TFPR increases if and only if the markup rises, and they attribute the increase in production efficiency following export to productivity improvements. We extend their work to allow for flexible returns to scale and show that it could be an important source of efficiency gains from shocks (such as export) that change firms’ market size. Flexible returns to scale also reconcile the facts that export increases markup but not TFPR.²

Because most datasets only record revenues without output prices or quantities, as in our main dataset, we develop a novel method to jointly estimate firm-level markup and productivity

¹Theoretical models have long posited that exporting enhances productivity. See [Clerides et al. \(1998\)](#) on learning-by-exporting, [Holmes and Schmitz Jr \(2001\)](#) on export-induced R&D, and [Melitz \(2003\)](#) on reallocation, for example. [De Loecker and Warzynski \(2012\)](#), [De Loecker et al. \(2016a\)](#), and [Garcia-Marin and Voigtländer \(2019\)](#) find that exports enhance productivity and markups.

²In their robustness check, [Garcia-Marin and Voigtländer \(2019\)](#) also noticed the possibility of increasing returns to scale and its potential impact on their main results. They argue that their main results are robust, although many industries exhibit increasing returns to scale in their robustness check.

based on revenue data, allowing for flexible returns to scale. The new method builds on the insights of [Hall \(1988\)](#) and [De Loecker and Warzynski \(2012\)](#), which relate markup to revenue-input ratios via cost minimization and use the relationship to compute markup after estimating output elasticity using quantity data. Our innovation is to incorporate this relationship directly into production estimation to control for the unobserved markup in the revenue function. This extension allows us to consistently estimate markup, productivity, and returns to scale at the firm level using revenue data, but not output prices. The new method can also be viewed as an extension of [Klette and Griliches \(1996\)](#) to allow for flexible markup. The low data requirement and multi-dimensional gains enable the potential widespread use of our method to better understand how policy changes (or actions like exports) generate multi-channel gains through market expansion and increasing returns, in addition to productivity and markup, providing a more accurate assessment of the policy effect.

There are three major findings. First, export has a significant physical productivity premium, echoing [Garcia-Marin and Voigtländer \(2019\)](#) and [Li et al. \(2025\)](#) that use output quantity data. Export raises exporters' productivity by 0.8% in our main specifications, after correcting for unobserved output prices and markup. In contrast, export has a negligible effect on the traditionally estimated revenue-based productivity (0.2-0.3%), although export increases markup substantially. This differs from [Garcia-Marin and Voigtländer \(2019\)](#), which demonstrates that TFPR increases in markup proportionally under constant returns to scale. We show that the negligible export effect on TFPR (even with increased markup) is due to increasing returns to scale: with increasing returns to scale, producing more following export reduces the marginal production costs, which offsets the positive effect of productivity and markup, leading to an ambiguous effect of export on TFPR.³

Second, exports increase firms' total market demand in terms of both market size and markup. The mean firm-level demand effect implied by the estimated demand parameter ranges from about 8.2-15.4%, depending on whether instrumental variables are used. The increased market demand allows the firm to produce more and potentially charge a higher markup. Indeed, export increases firms' markup by 0.8 to 1 percentage points on average, which serves as one crucial channel for exporters to increase their profitability.

Third, after correcting for markup, production shows substantially increasing returns to scale. In all specifications, estimates of returns to scale are significantly above 1. Correcting markup is crucial for consistently estimating returns to scale. When estimating the revenue function

³Although [Garcia-Marin and Voigtländer \(2019\)](#) acknowledged that neglecting increasing returns to scale could bias estimates of export-related productivity gains, they did not treat increasing returns to scale as a direct source of efficiency gains from exporting.

without correcting for markup, the estimates display decreasing returns to scale across all specifications, as in the literature (e.g., [Yu, 2015](#)). The intuition is straightforward: the revenue-based returns to scale are normalized by markup that is usually greater than 1. If the revenue-based returns to scale are between 0.9 to 1 as estimated in the literature and the markup is around 1.2, the implied true returns to scale are already greater than 1. Our result echoes [Klette and Griliches \(1996\)](#), who also find increasing returns to scale after controlling for industry-level average markup. We improve it by allowing for firm heterogeneity in markup. This result is also consistent with [De Loecker et al. \(2016a\)](#), who find that 68% of their sample observations exhibit increasing returns to scale estimated using output quantity data. Increasing returns to scale provides a new channel of export premium: the increased output after export reduces production costs, improving the exporters’ efficiency. It also explains why detecting export premiums on TFPR is difficult.

The estimation results imply a large efficiency gain from export, by reducing the marginal cost of production by an average of 1.71-1.85%. This result echoes [De Loecker et al. \(2016a\)](#) and [Garcia-Marin and Voigtländer \(2019\)](#), which rely on output quantity data. Increasing returns to scale plays an important role in firms’ efficiency gains from exports. Our decomposition shows that increasing returns to scale account for nearly two-thirds of the efficiency gain after export, with the remaining one-third attributable to the export productivity premium.

The improved production efficiency and the larger market improve exporters’ profitability and consumer welfare. In our data, exporters’ profit increases by about 25%. This results from the increased markup and reduced production costs arising from export premium on productivity and increasing returns to scale. We calculate the changes in the exporters’ output prices based on the estimated markup and changes in marginal costs. We find that the exporters’ price declines by 0.71-0.85% on average, implying an improvement in consumer welfare.

We validate our results by using a smaller sub-sample with firm-level output quantity. The unobserved markup is no longer a problem when estimating the physical output production function using traditional approaches. We find consistent results. Exports increase the firm’s productivity by 0.4-0.5% and production shows increasing returns to scale (1.069-1.079). When estimating the model using our method on this same smaller single-product sample (pretending no output quantity data are available), we find consistent results for export’s productivity effect (0.5–0.6%) and returns to scale (1.038–1.040), after correcting for the potential positive relationship between input and output prices (as proxies of input and output quality). In contrast, the traditional revenue-based approach shows decreasing returns to scale and fails to capture the productivity gains from exports as expected. The results

show that our results based on the new estimation methods are robust.

Our results are robust when using alternative estimation approaches such as [Akerberg et al. \(2015\)](#). We also checked for a more flexible Translog production function, in which case both expenditure shares and the flexible output elasticities contribute to the changes in markup. In this case, we find an even higher export premium on productivity, up to 1.8% under the IV specification. Returns to scale are increasing for most firms and increase in firm size, which echoes [De Loecker et al. \(2016a\)](#)'s finding in the relationship between marginal cost and output quantity. All other results are qualitatively and quantitatively similar to our main results. We also estimate the model industry by industry and the export effect and IRS appear in most industries. Finally, we also show that our main results are robust in data from other countries, such as Colombia.

The paper first contributes to the literature on production estimation and markup estimation. It is known that unobserved heterogeneity in output prices and markup biases productivity estimates using revenue data (e.g., [Klette and Griliches, 1996](#); [Foster et al., 2008](#)). [Klette and Griliches \(1996\)](#) partially address this problem by replacing the output prices in the revenue function with the demand function and estimating the revenue production function, assuming common markups for all firms. [De Loecker \(2011\)](#) and [De Loecker and Warzynski \(2012\)](#) propose an approach to estimating the firm-level markup and productivity with the knowledge of output quantity data. The key insight is that, for cost-minimizing firms, markups are equal to the wedge between the output elasticity of a variable input and the input's expenditure share in sales. However, because quantity data are unavailable in most firm-level datasets, most applications of this approach use revenue as a proxy for output quantity. This may cause substantial bias in the estimates of markup and productivity ([Bond et al., 2021](#); [Doraszelski and Jaumandreu, 2024](#); [De Ridder et al., 2025](#)). [De Ridder et al. \(2025\)](#) further show that using revenue data, the De Loecker-Warzynski approach can recover the cross-firm dispersion and overtime changes in markup (but not the average level of markup) under some restrictions on the production function and perfect instruments. Still, they found a moderate correlation between revenue-based and quantity-based markup at 0.63, suggesting that the original De Loecker-Warzynski approach may not recover markup consistently using revenue data.

Our new method shows that the original insight of [De Loecker and Warzynski \(2012\)](#) can be adjusted to consistently and jointly estimate firm-level markups, productivity, and returns to scale using revenue data without output prices. We achieve this goal by using the cost minimization condition to control for unobserved markup directly when estimating the revenue production function.

The paper also contributes to the large, important literature on the gains from exports, providing a comprehensive perspective to interpret many findings in the literature. Our findings that exports enhance productivity and markups are consistent with [De Loecker and Warzynski \(2012\)](#), [De Loecker et al. \(2016a\)](#), and [Garcia-Marin and Voigtländer \(2019\)](#). Our methodological innovation ensures that such analysis can be conducted even when output quantity data are unavailable. More importantly, we identify increasing returns to scale as a crucial source of the export premium at the firm level, alongside productivity and markup effects. Although [Garcia-Marin and Voigtländer \(2019\)](#) acknowledged that neglecting increasing returns to scale could bias estimates of export-related productivity gains, they did not treat increasing returns to scale as a direct source of efficiency gains from exporting. Our evidence indicates that increasing returns to scale is a vital channel through which exports generate efficiency improvements: they account for nearly two-thirds of the reduction in production costs observed among Chinese firms during our sample period. We find consistent results in the Colombia dataset as well. Finally, we provide an explanation for why the broader literature often reports limited or mixed effects of exports on productivity when using revenue-based productivity measures.

Finally, the paper provides empirical support for the new growth theory ([Romer, 1986](#)) and new trade theory ([Krugman, 1980a, 1995](#)), whose results are based on the assumption of increasing returns to scale at the macro level. Empirically, [Romer \(1986\)](#) presented evidence on increasing returns to scale at the macro level and used it as the fundamental to explain long-term growth. However, this result is questioned by [Basu and Fernald \(1997\)](#), who show that the estimated returns to scale vary largely depending on different levels of aggregation and could even be decreasing using gross output value or value-added data. Moreover, most of the estimates using microdata find decreasing returns to scale without correcting for markup (e.g., [Yu, 2015](#); [Dai et al., 2016](#), using the same data), except for a few exceptions (e.g., [Klette and Griliches, 1996](#); [De Loecker et al., 2016a](#)). We show that this might be due to an unobserved markup, which contaminates the estimation of output elasticity and thus returns to scale using revenue data. We provide a new method to estimate firm-level markup, productivity, and returns to scale simultaneously using revenue data. We provide strong evidence that, after correcting the impact of markup, production shows strong increasing returns to scale, even at the firm level. This result supports the argument of new growth theory and new trade theory.

In the rest of the paper, Section 2 introduces the data and motivational facts. Section 3 develops a framework to jointly estimate firm-level markup, productivity, and returns to scale consistently based on revenue data. Section 4 presents the estimation results. Section 5

validates our results in a smaller sample with rare output quantity data at the firm level. In Section 6, we calculate the gains from export in firm profitability and consumer welfare and evaluate the relative importance of the multiple sources of export premium. Section 7 checks the robustness of our results, and Section 8 concludes.

2 Data and Motivation

2.1 Data and Summary Statistics

Our empirical analysis focuses on the impact of direct ordinary exporting in the Chinese manufacturing industry. The analysis mainly uses two datasets.

The first dataset is the Annual Survey of Industrial Enterprises (ASIE) from 1998 to 2008, collected by the National Bureau of Statistics of China (NBS). The dataset contains detailed input and output information of all State Owned Enterprises (SOE) and non-SOE firms whose annual sales are greater than or equal to 5 million RMB (around 0.604 million U.S. dollars according to the exchange rate of 2000). The variables include total sales, material expenditure, capital stock, wage expenditure, labor employed, and other production information at the firm level. However, as in many other firm-level surveys, the dataset does not record these firms' output quantities or prices.

We clean the data using the following rules. First, we drop observations that contain negative or missing values for the following variables: total sales, total revenue, total employment, fixed capital, material expenditure, the cost of goods sold, city code, and holdings, following [Feenstra et al. \(2014\)](#) and [Dai et al. \(2016\)](#). Second, we only keep observations whose liquid assets are less than or equal to total assets, whose total fixed assets are less than or equal to total assets, whose net value of fixed assets is less than or equal to total assets, and whose labor employed is greater than or equal to eight, as in [Mo et al. \(2021\)](#). Moreover, we winsorize the data using P1 and P99 as the cutoff for major variables used in the analysis (i.e., total sales, labor employed, fixed capital, material expenditure, labor expenditure, and depreciation of the year).

The second dataset is China's custom records from 2000 to 2006 collected by China's General Administration of Customs (GAC). This dataset contains transaction-level export information, including trade type, price, quantity, etc. The trade type allows us to separate ordinary trade from processing trade, and we focus on ordinary exports.

We merge these two datasets following [Yu \(2015\)](#). The matched sample covers 15.28% of the observations (17.02% of firms) from the NBS dataset. Finally, our sample contains 1,234,292

observations from 401,020 firms, among which 188,633 observations from 68,267 firms were involved in direct ordinary export. This is highly consistent with those papers focusing on the same datasets and adopting similar merging methods. ⁴

The matched sample is our main dataset for the empirical analysis. Table 1 reports the summary statistics of the main variables for our empirical analysis. We also calculate a measure of “raw markup ($\tilde{\mu}_{jt}$)”, defined as the ratio of total sales to the sum of material expenditure and wage expenditure. We show in Appendix A.1 that the true markup (μ_{jt}), defined as the price-marginal cost ratio, equals the raw markup ($\tilde{\mu}_{jt}$) normalized by returns to scale. Hence, the raw markup is an important source of variation for true markup. In the special case with constant output elasticities (e.g., Cobb-Douglas production function), the raw markup is the only source of variation in the true markup. The insight is similar to De Loecker (2011) and De Loecker and Warzynski (2012).

Table 1: Summary Statistics

Statistics	Median	Mean	SD	IQR	IDR
Total Sales (million USD)	1.977	4.456	7.357	3.575	9.742
Material Expenditure (million USD)	1.546	3.506	5.866	2.806	7.663
Capital Stock (million USD)	0.442	1.392	2.927	1.066	3.296
Wage Expenditure (million USD)	0.121	0.244	0.371	0.205	0.534
Labor Employed	101	183.083	241.282	152	380
Raw Markup ($\tilde{\mu}$)	1.154	1.182	0.239	0.228	0.470
Export Probability	0	0.153	0.335	0	1
Export Share (Full sample)	0	0.067	0.215	0	0.153
Export Share (Con. on Export)	0.342	0.440	0.373	0.753	0.986
Number of Exporting Firms	68,267 (17.02%)				
Total Number of Firms	401,020				
Exporting Observations	188,633 (15.28%)				
Observations	1,234,292				

Note: All monetary values here are in millions of 2000 U.S. dollars.

Besides the main datasets, we motivate this study and validate our new methods using a smaller sample of firms that have output quantity data. The output quantity survey from 2000 to 2006 contains measures of output quantity at the firm-product level, allowing us to calculate a rough measure of firm output prices. After merging it with our main data, we have 444,475 observations with output quantity. Among them, 88,487 observations were from exporting firms, accounting for a bit over one-third of the 243,472 exporting observations in

⁴ Dai et al. (2016) reported 1,244,382 observations from 424,546 firms with 68,865 direct exporters, and Ge et al. (2015) reported 77,087 direct exporters in their merged datasets respectively.

our main sample. Although we did not use the smaller dataset in our main analysis due to its incomplete coverage, it provides good motivation for our study and validation of our new estimation method.

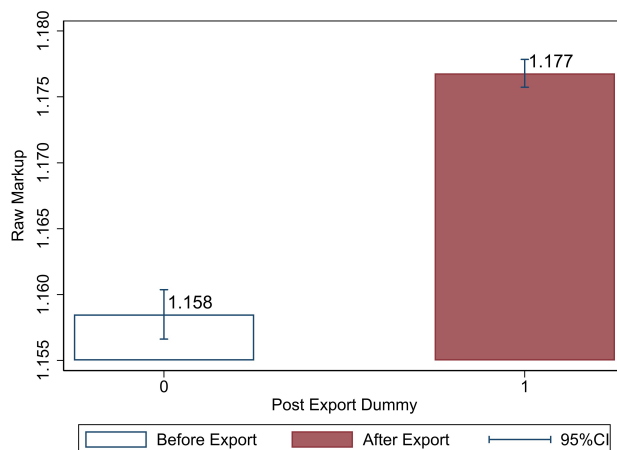
2.2 Motivational Facts

As one of the most important events in the Chinese economy in the 21st century so far, China's accession to the WTO in 2001 led massive Chinese firms to export. Did Chinese exporters enjoy efficiency gains from exports? This subsection presents two motivational facts that strongly support large efficiency gains from export: a large export premium in markup and a substantial reduction of output prices after export.

2.2.1 Markup Premium of Export

Although firm-level markup is not directly observed in the data, we use the revenue-to-variable cost ratio (raw markup) as a proxy for markup. In the special case of the Cobb-Douglas production function, the raw markup is the only source of variation in true markup. Table 1 shows that there is a large heterogeneity in markup across firms, with an interquartile range of 0.228. Figure 1 compares the average markup of firms before and after they start exporting. It is shown that the exporters' markup is significantly higher after they start exporting than before. The margin is about 2 percentage points.

Figure 1: Raw Markup Mean Before & After Exporting



Note: This figure plots the average markup of new exporters in the years before and after they start exporting, with the 95% confidence interval.

2.2.2 Price Decline after Export

Although output quantity and prices are not available in our main data, we can leverage the smaller output-quantity survey described above to construct firm-level output prices for a subset of firms. This allows us to examine the impact of export on firms' output prices.⁵ We estimate the effect of export on the prices charged by exporters, after controlling for firm fixed effects, year fixed effects, and a series of firm characteristics:

$$AveragePrice_{jt} = \beta_0^P + \beta_{exp}^P D_{jt}^{exp} + \beta^X \mathbf{X}_{jt} + \gamma_j + \gamma_t + \xi_{jt}^P, \quad (1)$$

where D_{jt}^{exp} is the export dummy equaling 1 if firm j is exporting at time t and 0 otherwise; the coefficient of interest, β_{exp}^P , captures the average export effect on firm-level prices. $\mathbf{X}_{jt}$ is a set of control variables, including firm size (total sales) and capital intensity. γ_j and γ_t represent firm and year fixed effects, respectively, and ξ_{jt}^P is an i.i.d. shock to the firm's average price.

To minimize any composition effect between exporters and non-exporters, we further restrict the sample to firms that start exporting at some point during our sample period. Under this restriction, β_{exp}^P identifies the within-firm change in price before and after the firm starts exporting, rather than any level difference between exporters and non-exporters. The estimation results are reported in Table 2. Exporting reduces the prices charged by exporters significantly, consistent with the findings in Li et al. (2025). As shown in Column (2), exporting reduces price by 3.7% on average. The price change in level terms in Column (1) goes in the same direction.

The lowered output prices, together with the increased markup as shown above, imply an export premium on production efficiency: exporting reduces marginal production costs, so that the firm can charge a higher markup at a lower price. The gain in production efficiency may arise from an export productivity premium or increased scale returns (if any) given the larger market after export. However, if the markup is greater than 1 and unobserved, it will bias down the estimate of returns to scale and its resulting export premium based on the revenue production function. In Section 3, we develop a new method to estimate the multiple

⁵To keep the firm-level price well defined, we restrict attention to single-product firms and define the firm-level output price as the ratio of total sales (from the full ASIE sample) to output quantity reported in the quantity survey. Restricting the sample to single-product firms is necessary because, for multi-product firms, we observe quantities at the firm-product level but revenues only at the firm level. Hence, constructing a firm-level price index for multi-product firms would require additional assumptions about the within-firm product mix. We also considered two alternative measures that include multi-product firms: defining the firm-level price as total sales divided by either the simple sum of reported quantities across products or the reported quantity of the firm's largest product. Both approaches yield quantitatively similar results. For transparency, we report only the cleanest specification, based on single-product firms.

Table 2: Export and Firms' Average Price Change

Parameter	(1) Average Price	(2) Average Price (log)
D_{jt}^{exp}	-12.449* (6.9448)	-0.037*** (0.0103)
Firm Size (Sales)	YES	YES
Capital Intensity	YES	YES
Firm FE	YES	YES
Year FE	YES	YES
Adjusted R^2	0.715	0.937
Observations	69,367	69,367

Note 1: The sample is constructed from the subset of firms reporting output-quantity information described in Section 5, restricted to single-product firms so that firm-level prices are well defined. We further restrict the sample to firms that start exporting during our sample period.

Note 2: The dependent variable is the firm-level average price, computed as the firm's total sales revenue reported in the full ASIE sample divided by its reported output quantity. The coefficient on D_{jt}^{exp} therefore captures the within-firm change in price before and after the firm starts exporting, rather than any level difference between exporters and non-exporters. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sources of gains from exports, using widely available revenue data and input expenditure.

2.3 Key Insights: Increasing Returns to Scale and TFPR

Why may TFPR fail to capture the efficiency gains from exports? To provide insight, we decompose TFPR into its contributing factors. By definition, changes in TFPR are contributed by changes in physical productivity and prices,

$$\Delta \text{TFPR} = \Delta p + \Delta \omega = \Delta \ln \mu + \Delta mc + \Delta \omega. \quad (2)$$

Where $p, \omega, \ln \mu$, and mc represent the logarithm of output price, physical productivity, markup, and marginal costs of production, respectively. ΔX represents changes in X . The second equality in (2) holds following the definition of markup, $\ln \mu = p - mc$. In the special case of constant returns to scale, a 1-percent increase in productivity leads to a 1-percent decline in marginal costs, offsetting each other. Therefore, TFPR changes if and only if there is a change in markup. [Garcia-Marin and Voigtländer \(2019\)](#) provide empirical support using firm-level data from Colombia and Chile.

However, [Garcia-Marin and Voigtländer \(2019\)](#) may not be the whole story, because, while export increases markup as shown above, its effect on TFPR is negligible, as will be shown in Section 4 using the Chinese data. To rationalize this observation, we generalize [Garcia-Marin and Voigtländer \(2019\)](#) to allow for non-constant returns to scale. In this case, (2) can be

written as:⁶

$$\begin{aligned}
\Delta \text{TFPR} &= \Delta p + \Delta \omega = \Delta \ln \mu + \Delta mc + \Delta \omega \\
&= \Delta \ln \mu + \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} \Delta y - \frac{1}{\alpha_L + \alpha_M} \Delta \omega + \Delta \omega \\
&= \Delta \ln \mu + \left(1 - \frac{1}{\alpha_L + \alpha_M}\right) (\Delta \omega - \Delta y),
\end{aligned} \tag{3}$$

where y represents output quantity (in logarithm) and (α_L, α_M) represent output elasticities of labor and material, respectively. (3) shows how returns to scale matter. With increasing returns to scale ($\alpha_L + \alpha_M > 1$), an increase in physical productivity (ω) will increase TFPR conditional on output level. This is because, with increasing returns to scale, a 1-percent increase in productivity reduces marginal costs by less than 1 percent, conditional on output level. More importantly, producing more after starting export reduces TFPR through increasing returns to scale, which may offset the potentially positive effect of export on markup and productivity. As a result, even if exports increase markup and productivity, the effect on TFPR may still be ambiguous. This explains why the literature may fail to find an export premium based on TFPR, even when markup increases following export. Notably, [Garcia-Marin and Voigtländer \(2019\)](#) recognized that ignoring increasing returns to scale may bias estimates of export-related productivity gains; however, they did not treat increasing returns to scale as a direct source of efficiency gains from exporting.

The remaining question is: Does technology show increasing returns to scale at the firm level? In the following sections, we show that traditional production function estimation approaches based on revenue data underestimate returns to scale. After controlling for firm heterogeneity in markup, firm-level technology shows strong increasing returns to scale, and it is an important source of gains from export in addition to productivity and markup.

3 Empirical Model

Because output prices are not reported in our main data as in most micro datasets, we developed a new method to consistently estimate firm-level markup, productivity, and returns to scale jointly, using the widely available revenue data but not output prices. The method also allows us to estimate the multiple sources of gains from exports. The key idea is to use the variation in revenue-to-total variable cost ratios to control for markup, as implied by the cost minimization conditions, sharing a similar insight as [De Loecker and Warzynski \(2012\)](#). As an obvious advantage, while the latter requires output quantity data to estimate

⁶Please refer to Appendix A.7 for the detailed derivations.

output elasticities first and then use them to construct firm-level markup, our new method does not require output quantity or prices. Instead, we use the cost minimization conditions to directly control for unobserved markup and estimate the markup and output elasticities jointly by expanding [Klette and Griliches \(1996\)](#) to allow for firm heterogeneity in markup.

3.1 Setup

Firms produce heterogeneous but single products, and engage in monopolistic competition in the market.

Production. A firm produces a single output in physical unit Y_{jt} using labor (L_{jt}), intermediate input (M_{jt}), and capital (K_{jt}) as inputs, summarized in the input vector $\mathbf{X}_{jt}^I = (L_{jt}, M_{jt}, K_{jt})$. The gross production function of firm j at time t is Cobb-Douglas:

$$Y_{jt} = e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K}, \quad (4)$$

where $\theta = (\alpha_L, \alpha_M, \alpha_K)$ is the vector of the corresponding output elasticities to be estimated. ω_{jt} is the physical total factor productivity (TFPQ) observed by the firm before production. Note that we assume firms observe their productivity fully and load the unobserved revenue shifter into the demand function, as discussed below. This ensures a noise-free relationship between markup and revenue-to-variable cost ratios, which is used to control for markup, while retaining unobserved idiosyncratic shocks in revenue functions. Of course, firm productivity is subject to i.i.d. shocks each period. It evolves as a first-order Markov process with exporting as a shifter:

$$\omega_{jt} = \beta_0^\omega + \beta_1^\omega \omega_{jt-1} + \beta_{exp}^\omega D_{jt}^{exp} + \epsilon_{jt}^\omega, \quad (5)$$

where D_{jt}^{exp} is a dummy variable indicating whether firm j is involved in export business at time t , as defined in (1). β_{exp}^ω captures the immediate exporting premium on productivity, and ϵ_{jt}^ω is the i.i.d. innovation shock to firm j 's productivity. β_1^ω is the persistence parameter and β_0^ω is a constant.

Demand and Revenue Function. To separately identify TFPQ from demand-side shocks, we model the residual demand facing firm j at time t following the demand-appeal framework similar to the insight of [Feenstra and Romalis \(2014\)](#). The inverse demand function is:

$$\frac{P_{jt}/e^{z_{jt}}}{\bar{P}_{Jt}} = \left(\frac{Y_{jt}e^{z_{jt}}}{\bar{Y}_{Jt}} \right)^{1/\eta_{jt}}, \quad (6)$$

where $\bar{Y}_{jt}$ and $\bar{P}_{jt}$ are industry-level output and price indices, respectively, and $\eta_{jt} < -1$ is the firm-time-specific price elasticity of demand. $z_{jt} = \beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D$ is the product appeal, which summarizes the impact of market size, consumer subjective evaluation, and product quality on demand. It shifts with export participation D_{jt}^{exp} . Given the demand appeal, consumers care about the effective quantity $\tilde{Y}_{jt} = Y_{jt} e^{z_{jt}}$ rather than the physical quantity, and face an effective price $\tilde{P}_{jt} = P_{jt} / e^{z_{jt}}$. Therefore, the demand function in (6) is appeal-adjusted. ϵ_{jt}^D represents an unforecastable i.i.d. demand shock that is not observed by the firm until the production is completed. This specification serves as a robust first-order log-linear approximation of a generalized VES preference system (Kimball, 1995) and, equivalently, can be derived exactly from a nested CES framework with oligopolistic competition (Atkeson and Burstein, 2008), in which a firm's perceived demand elasticity depends on its market share.

Taking the natural logarithm of (6) and rearranging yields the log-linear demand function:

$$y_{jt} = \eta_{jt} p_{jt} + \varphi_{jt}, \quad (7)$$

where the composite demand shifter is $\varphi_{jt} \equiv \bar{y}_{jt} - \eta_{jt} \bar{p}_{jt} - (1 + \eta_{jt}) z_{jt}$. The demand function is characterized by the demand elasticity η_{jt} and the demand shifter φ_{jt} , both of which vary across firms and over time, so the model allows for firm heterogeneity in markup and market size. With the standard profit-maximizing condition for monopolistic competition, the firm-level markup equals $\mu_{jt} \equiv \eta_{jt} / (1 + \eta_{jt})$.⁷ Combining (7) with the production function (4) and exploiting the algebraic identity $1 - 1/\mu_{jt} = -1/\eta_{jt}$, the (log) revenue production function can be rewritten elegantly as:

$$r_{jt} - \bar{r}_{jt} = \frac{1}{\mu_{jt}} (\alpha_L l_{jt} + \alpha_M m_{jt} + \alpha_K k_{jt} + \omega_{jt} - \bar{y}_{jt} + \beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D), \quad (8)$$

where $\bar{r}_{jt} \equiv \bar{p}_{jt} + \bar{y}_{jt}$ is the industry-level (log) revenue index and l_{jt} , m_{jt} , k_{jt} are the logarithms of L_{jt} , M_{jt} , K_{jt} . Equation (8) extends Klette and Griliches (1996) to the more general case with firm heterogeneity in markup and a structurally interpretable demand shock.

Three lessons follow from this equation. First, if there is markup ($\mu_{jt} > 1$), the output elasticities and, as a result, returns to scale will be understated if we estimate the revenue production function as a proxy of the physical production function. Second, the revenue productivity, typically defined as $\frac{1}{\mu_{jt}}(\omega_{jt} + z_{jt})$ in the literature, can bias the estimate of the

⁷In Appendix A.2, we formally prove that defining demand over effective quantities and prices preserves the standard profit-maximizing markup condition $\mu_{jt} = \eta_{jt} / (1 + \eta_{jt})$, ensuring the validity of our subsequent structural substitutions.

policy effect on productivity, as it conflates the effects on productivity, markup, and demand factors. Third, firm heterogeneity in markup interacts non-linearly with inputs and ex-post demand shocks; if unobserved, this will bias the estimates of (8). It is also difficult to find valid instrumental variables for markup, because markup correlates with inputs, outputs, and market conditions.⁸

3.2 Joint Estimation of Markup and Production Function

We develop a new approach to consistently estimate (8) for productivity, markup, and returns to scale, using the widely available revenue data. Our strategy is to find a reliable proxy for markup and control it directly in the production estimation stage. Specifically, we use the revenue-to-variable-cost ratio to control for firm heterogeneity in markups. This proxy follows from firms' cost-minimization conditions in the spirit of [De Loecker and Warzynski \(2012\)](#).

Cost Minimization and the Markup Proxy. Assume that firms choose labor and material to minimize production cost, given productivity and quasi-fixed capital stock. Denote $\mathbf{X}_{jt}^V = (L_{jt}, M_{jt})$ as the vector of variable inputs and $\mathbf{P}_{jt}^V = (P_{jt}^L, P_{jt}^M)$ the corresponding variable input prices. The associated Lagrangian is:

$$\mathcal{L}(\mathbf{X}_{jt}^V) = \mathbf{P}_{jt}^V \cdot \mathbf{X}_{jt}^V + \lambda_{jt} (Y_{jt} - e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K}), \quad (9)$$

in which the Lagrange multiplier λ_{jt} represents the marginal cost of production. Combining the first-order conditions for labor and material yields the structural relationship between markup, returns to scale, and the revenue-to-variable-cost ratio:

$$\mu_{jt} = (\alpha_L + \alpha_M) \cdot \frac{R_{jt}}{P_{jt}^L L_{jt} + P_{jt}^M M_{jt}} = S^V \cdot \tilde{\mu}_{jt}, \quad (10)$$

where $S^V \equiv \alpha_L + \alpha_M$ is the sum of output elasticities of variable inputs (returns to scale of variable inputs, RTSV) and $\tilde{\mu}_{jt} \equiv R_{jt}/(P_{jt}^L L_{jt} + P_{jt}^M M_{jt})$ is the observable raw markup. Equation (10) shows that markup is known from the raw markup, up to RTSV.⁹ Importantly, equation (10) is guaranteed by the accounting identity from cost minimization, and the firm does not require any output-price information when making cost-minimizing variable-input choices. Because the idiosyncratic demand shock ϵ_{jt}^D is realized ex-post and lies outside the

⁸Relevant discussion can also be found in [Bond et al. \(2021\)](#) and [Doraszelski and Jaumandreu \(2024\)](#). [Doraszelski and Jaumandreu \(2019\)](#) also discusses the problem of omitting the demand heterogeneity when estimating the production function using the control function approach, even when quantity output data is available.

⁹The relationship among the three terms in (10) is general: it holds for any form of production function and demand function, in which case the RTSV may be flexible and vary across firms and over time.

firm's information set when input choices are made, it is strictly orthogonal to inputs and to $\tilde{\mu}_{jt}$ conditional on the ex-ante state variables. Our strategy is to use this equation to control for unobserved markup when estimating the revenue function (8).

Markup-Adjusted Revenue Function. A major challenge in estimating (8) is that the unobserved markup μ_{jt} interacts non-linearly with both inputs and unobserved shocks. We solve this problem by controlling for the unobserved markup μ_{jt} using (10). By replacing (10) into (8) and multiplying both sides by the observable term $\tilde{\mu}_{jt}$, we derive the markup-adjusted revenue function after some rearrangements:

$$\begin{aligned} \tilde{\mu}_{jt} \cdot (r_{jt} - \bar{r}_{Jt}) = & \frac{\alpha_L}{\alpha_L + \alpha_M} l_{jt} + \frac{\alpha_M}{\alpha_L + \alpha_M} m_{jt} + \frac{\alpha_K}{\alpha_L + \alpha_M} k_{jt} + \frac{1}{\alpha_L + \alpha_M} \omega_{jt} \\ & - \frac{1}{\alpha_L + \alpha_M} \bar{y}_{Jt} + \frac{\beta_{exp}^D}{\alpha_L + \alpha_M} D_{jt}^{exp} + \frac{1}{\alpha_L + \alpha_M} \epsilon_{jt}^D, \end{aligned} \quad (11)$$

in which the only remaining stochastic component on the right-hand side is the strictly additive ex-post shock ϵ_{jt}^D with the constant coefficient $1/(\alpha_L + \alpha_M)$. Because the raw markup $\tilde{\mu}_{jt}$ is observed in the data, the left-hand side is just data. This equation forms our main estimation equation, and it can be estimated using standard approaches such as [Levinsohn and Petrin \(2003\)](#) and [Akerberg \(2016\)](#). We use [Levinsohn and Petrin \(2003\)](#) in our main results, but the results are robust when using alternative methods, as shown in the robustness check. The only minor difference is that because α_L and α_M are not separately identified from variations in l_{jt} and m_{jt} , we exploit variation in the industry-average output-quantity index $\bar{y}_{Jt}$ to identify the returns to scale $\alpha_L + \alpha_M$ as in [Klette and Griliches \(1996\)](#).

Control-Function Strategy. Given the timing assumption, material demand is independent of the ex-post demand shock ϵ_{jt}^D . The firm's optimal material demand depends only on the observed state variables $(\omega_{jt}, k_{jt}, D_{jt}^{exp}, \bar{y}_{Jt})$ under LP's estimation procedure¹⁰:

$$m_{jt} = m(\omega_{jt}, k_{jt}, D_{jt}^{exp}, \bar{y}_{Jt}).$$

Material demand is strictly increasing in ω_{jt} under the monotonicity assumption, so we can invert it to non-parametrically control for productivity, $\omega_{jt} = m^{-1}(m_{jt}, k_{jt}, D_{jt}^{exp}, \bar{y}_{Jt})$. Using it to replace the productivity term ω_{jt} in (11), we derive the following stage one estimation equation in the production estimation procedure:

$$\tilde{\mu}_{jt} \cdot (r_{jt} - \bar{r}_{Jt}) = \frac{\alpha_L}{\alpha_L + \alpha_M} l_{jt} + \phi_{jt}(m_{jt}, k_{jt}, \bar{y}_{Jt}, D_{jt}^{exp}) + \frac{1}{\alpha_L + \alpha_M} \epsilon_{jt}^D, \quad (12)$$

¹⁰Alternatively, in the ACF approach the state variables are $(l_{jt}, \omega_{jt}, k_{jt}, D_{jt}^{exp}, \bar{y}_{Jt})$. We use LP as the main estimation procedure and retain ACF as a robustness check.

where $\phi_{jt}(m_{jt}, k_{jt}, \bar{y}_{Jt}, D_{jt}^{exp})$ is a non-parametric function of m_{jt} , k_{jt} , $\bar{y}_{Jt}$, and D_{jt}^{exp} implied by the model, approximated by a cubic polynomial. This equation can be estimated using the nonlinear least squares method. By estimating (12), we identify the coefficient on l_{jt} , $\hat{\alpha} \equiv \alpha_L/(\alpha_L + \alpha_M)$, the residual $\epsilon_{jt}^D/(\alpha_L + \alpha_M)$, and the non-parametric function $\hat{\phi}_{jt}$.

By definition, $\hat{\phi}_{jt} = \frac{\alpha_M}{\alpha_L + \alpha_M} m_{jt} + \frac{\alpha_K}{\alpha_L + \alpha_M} k_{jt} + \frac{1}{\alpha_L + \alpha_M} \omega_{jt} - \frac{1}{\alpha_L + \alpha_M} \bar{y}_{Jt} + \frac{\beta_{exp}^D}{\alpha_L + \alpha_M} D_{jt}^{exp}$ from (11) and (12). We then solve productivity, ω_{jt} , as a function of the estimated $\hat{\phi}_{jt}$, observed variables, and parameters as follows:

$$\omega_{jt} = (\alpha_L + \alpha_M) \hat{\phi}_{jt} - \alpha_M m_{jt} - \alpha_K k_{jt} + \bar{y}_{Jt} - \beta_{exp}^D D_{jt}^{exp}. \quad (13)$$

Substituting (13) into the productivity-evolution process (5) for ω_{jt} and ω_{jt-1} yields:

$$\begin{aligned} (\alpha_L + \alpha_M) \hat{\phi}_{jt} &= \alpha_M m_{jt} + \alpha_K k_{jt} - \bar{y}_{Jt} + \beta_{exp}^D D_{jt}^{exp} + \beta_0^\omega \\ &+ \beta_1^\omega \left[(\alpha_L + \alpha_M) \hat{\phi}_{jt-1} - \alpha_M m_{jt-1} - \alpha_K k_{jt-1} + \bar{y}_{Jt-1} - \beta_{exp}^D D_{jt-1}^{exp} \right] \\ &+ \beta_{exp}^\omega D_{jt}^{exp} + \epsilon_{jt}^\omega. \end{aligned} \quad (14)$$

In the second stage, we estimate (14) using the Generalized Method of Moments (GMM). The details can be found in appendix A.4. After the estimation, we calculate the markup μ_{jt} via (10).

Endogeneity of Export Status. Because export may be correlated with productivity innovation, directly estimating (14) is subject to the standard endogeneity problem due to firms' self-selection into export markets. Following the standard exercise in the international-trade literature (e.g., [Pierce and Schott, 2016](#); [Handley and Limão, 2017](#); [Alessandria et al., 2025](#)), we instrument firms' export status by the interaction of the industry-level Normal Trade Relations (NTR) tariff gap that the U.S. government charges to Chinese exporters ($NTRGap_{i,t=1999}$), the WTO accession dummy ($D_t^{PostWTO}$), and the lagged firm's export status ($D_{j,t-1}^{exp}$). Specifically, we use the triple interaction $NTRGap_{i,t=1999} \times D_t^{PostWTO} \times D_{j,t-1}^{exp}$, controlling for $NTRGap_{i,t=1999} \times D_t^{PostWTO}$, $NTRGap_{i,t=1999} \times D_{j,t-1}^{exp}$, and other state variables (e.g., firms' productivity entering through the control function), to instrument firms' current export status.

This instrumental-variable strategy exploits the exogenous reduction in trade-policy uncertainty following China's PNTR status to identify the causal effect of exporting. The instrument's relevance is grounded in the theoretical framework of [Pierce and Schott \(2016\)](#) and [Handley and Limão \(2017\)](#): the granting of PNTR permanently eliminated the threat of tariff hikes—measured by $NTRGap$ —thereby disproportionately incentivizing firms in high-

gap industries, particularly those with prior export exposure ($D_{j,t-1}^{exp}$), to solidify their U.S. market presence. Crucially, the instrument satisfies exogeneity because the cross-sectional variation in $NTRGap$ is primarily driven by the non-NTR rates set by the U.S. Smoot–Hawley Tariff Act of 1930; as corroborated by [Alessandria et al. \(2025\)](#), these historical rates were determined seven decades prior to our sample period and are thus plausibly orthogonal to unobserved contemporary supply or demand shocks affecting Chinese firms in the 2000s.

Timing Assumption and Identification. Our empirical framework adopts an “*ex-ante* production, *ex-post* pricing” timing: firm j chooses flexible inputs (L_{jt}, M_{jt}) and produces a fixed physical output Y_{jt} based on ω_{jt} , D_{jt}^{exp} , capital, and expectations for future; the idiosyncratic demand shock ϵ_{jt}^D is realized after the production is completed, which then determines the market-clearing price P_{jt} . This timing delivers three properties that simplify the identification. First, it ensures scalar unobservability in the productivity proxy function. Material demand depends on a single unobservable ω_{jt} , because ϵ_{jt}^D has not yet been drawn at the input-choice stage. Second, it ensures strict monotonicity. $\partial m_{jt}/\partial \omega_{jt} > 0$ makes the control-function inversion valid. Third, it ensures strict exogeneity: multiplying (8) by the observable raw markup $\tilde{\mu}_{jt}$ algebraically cancels the nonlinear interactions between μ_{jt} and unobservables, leaving $\epsilon_{jt}^D/(\alpha_L + \alpha_M)$ as a strictly additive residual orthogonal to *ex-ante* inputs. These properties simultaneously circumvent the two prominent critiques of the production approach discussed in [Doraszelski and Jaumandreu \(2024\)](#) and [Bond et al. \(2021\)](#). The former shows that the standard Olley–Pakes/Levinsohn–Petrin inversion fails when firms observe demand shocks before choosing inputs (a two-unobservables problem ruled out here by our timing). The latter shows that using the revenue elasticity in a [De Loecker and Warzynski \(2012\)](#)-style ratio estimator mechanically forces $\mu_{jt} \equiv 1$ (which our specification avoids by using $\tilde{\mu}_{jt}$ to transform the dependent variable rather than estimating a revenue production function). Appendix [A.3](#) develops these arguments in detail.

3.3 Extension: Translog Production Function

One limitation of the Cobb–Douglas production function is that the output elasticities are constant. Hence, the variation of markup only comes from that of raw markup. This subsection shows that our empirical framework extends naturally to more general production functions. We consider the translog production function as an example:

$$\begin{aligned} y_{jt} = & \alpha_L l_{jt} + \alpha_M m_{jt} + \alpha_K k_{jt} + \alpha_{LL} l_{jt} l_{jt} + \alpha_{MM} m_{jt} m_{jt} + \alpha_{KK} k_{jt} k_{jt} \\ & + \alpha_{LM} l_{jt} m_{jt} + \alpha_{LK} l_{jt} k_{jt} + \alpha_{MK} m_{jt} k_{jt} + \omega_{jt}. \end{aligned} \quad (15)$$

Through a procedure analogous to that used to derive (8), we obtain the translog-form revenue production function. After substituting the markup using (10) and the structural demand shifter $\varphi_{jt} \equiv \bar{y}_{Jt} - \eta_{jt}\bar{p}_{Jt} - (1 + \eta_{jt})z_{jt}$, the markup-adjusted estimation equation reads:

$$\begin{aligned} \tilde{\mu}_{jt} \cdot (r_{jt} - \bar{r}_{Jt}) = & \frac{\alpha_L}{\alpha_L^* + \alpha_M^*} l_{jt} + \frac{\alpha_M}{\alpha_L^* + \alpha_M^*} m_{jt} + \frac{\alpha_K}{\alpha_L^* + \alpha_M^*} k_{jt} + \frac{\alpha_{LL}}{\alpha_L^* + \alpha_M^*} l_{jt}^2 \\ & + \frac{\alpha_{MM}}{\alpha_L^* + \alpha_M^*} m_{jt}^2 + \frac{\alpha_{KK}}{\alpha_L^* + \alpha_M^*} k_{jt}^2 + \frac{\alpha_{LM}}{\alpha_L^* + \alpha_M^*} l_{jt} m_{jt} + \frac{\alpha_{LK}}{\alpha_L^* + \alpha_M^*} l_{jt} k_{jt} \\ & + \frac{\alpha_{MK}}{\alpha_L^* + \alpha_M^*} m_{jt} k_{jt} + \frac{1}{\alpha_L^* + \alpha_M^*} \omega_{jt} - \frac{1}{\alpha_L^* + \alpha_M^*} \bar{y}_{Jt} \\ & + \frac{\beta_{exp}^D}{\alpha_L^* + \alpha_M^*} D_{jt}^{exp} + \frac{1}{\alpha_L^* + \alpha_M^*} \epsilon_{jt}^D, \end{aligned} \quad (16)$$

where the variable output elasticities α_X^* , $X \in \{L, M, K\}$, are given by $\partial y_{jt} / \partial x_{jt}$ and are functions of l_{jt} , m_{jt} , and k_{jt} .¹¹ Hence, the output elasticities vary across firms and over time. As in the Cobb-Douglas case, the linearization through $\tilde{\mu}_{jt}$ leaves only the strictly additive ex-post shock $\epsilon_{jt}^D / (\alpha_L^* + \alpha_M^*)$ on the right-hand side, eliminating the non-linear interaction $\tilde{\mu}_{jt} \beta_{exp}^D D_{jt}^{exp}$ that would otherwise appear under alternative timing assumptions. Because we cannot separate the error term and the RTSV $(\alpha_L^* + \alpha_M^*)$, which depends on the input usage l_{jt} , m_{jt} , and k_{jt} , OLS is no longer suitable for estimating (16). We therefore use a GMM procedure in the first step. Specifically, from (16) we express the demand-shock term ϵ_{jt}^D as a function of observed data and the unobserved productivity.¹² By proxying the unobserved productivity using the control function as in the Cobb-Douglas case, we can rewrite the residual as $\epsilon_{jt}^D = \bar{y}_{Jt} - \Phi_{jt}(\tilde{\mu}_{jt}(r_{jt} - \bar{r}_{Jt}), l_{jt}, m_{jt}, k_{jt}, D_{jt}^{exp})$, where Φ_{jt} is a cubic polynomial of $\tilde{\mu}_{jt}(r_{jt} - \bar{r}_{Jt})$, l_{jt} , m_{jt} , k_{jt} , and D_{jt}^{exp} . Because ϵ_{jt}^D is i.i.d. and, by the ex-ante/ex-post timing assumption, uncorrelated with firm production decisions, we estimate the model using the following moment conditions:

$$E \left(\epsilon_{jt}^D (\beta^\Phi) \begin{pmatrix} (\tilde{\mu}_{j,t-1} (r_{j,t-1} - \bar{r}_{J,t-1}))^a l_{jt}^b m_{jt}^c k_{jt}^d \\ D_{jt}^{exp} \end{pmatrix} \right) = 0 \quad (17)$$

where $a + b + c + d \in \{0, 1, 2\}$.

¹¹Specifically, $\alpha_L^* \equiv \alpha_L + 2\alpha_{LL}l_{jt} + \alpha_{LM}m_{jt} + \alpha_{LK}k_{jt}$, $\alpha_M^* \equiv \alpha_M + 2\alpha_{MM}m_{jt} + \alpha_{LM}l_{jt} + \alpha_{MK}k_{jt}$, $\alpha_K^* \equiv \alpha_K + 2\alpha_{KK}k_{jt} + \alpha_{LK}l_{jt} + \alpha_{MK}m_{jt}$.

¹² $\epsilon_{jt}^D = (\alpha_L^* + \alpha_M^*) \cdot \tilde{\mu}_{jt}(r_{jt} - \bar{r}_{Jt}) - (\alpha_L l_{jt} + \alpha_M m_{jt} + \alpha_K k_{jt} + \alpha_{LL} l_{jt}^2 + \alpha_{MM} m_{jt}^2 + \alpha_{KK} k_{jt}^2 + \alpha_{LM} l_{jt} m_{jt} + \alpha_{LK} l_{jt} k_{jt} + \alpha_{MK} m_{jt} k_{jt} + \omega_{jt}) + \bar{y}_{Jt} - \beta_{exp}^D D_{jt}^{exp}$.

β^Φ collects the parameters of $\Phi(\cdot)$ to be estimated. Given the estimates of β^Φ and the residual $\hat{\epsilon}_{jt}^D$, productivity can be expressed as

$$\begin{aligned} \omega_{jt} = & (\alpha_L^* + \alpha_M^*) \cdot [\tilde{\mu}_{jt}(r_{jt} - \bar{r}_{Jt})] - (\alpha_L l_{jt} + \alpha_M m_{jt} + \alpha_K k_{jt} + \alpha_{LL} l_{jt}^2 + \alpha_{MM} m_{jt}^2 \\ & + \alpha_{KK} k_{jt}^2 + \alpha_{LM} l_{jt} m_{jt} + \alpha_{LK} l_{jt} k_{jt} + \alpha_{MK} m_{jt} k_{jt} - \bar{y}_{Jt} + \beta_{exp}^D D_{jt}^{exp} + \hat{\epsilon}_{jt}^D). \end{aligned} \quad (18)$$

Then, analogous to the second-stage procedure for the Cobb-Douglas case, we express the productivity innovation ϵ_{jt}^ω as a function of the parameters to be estimated, $\epsilon_{jt}^\omega(\alpha_L, \alpha_M, \alpha_K, \alpha_{LL}, \alpha_{MM}, \alpha_{KK}, \alpha_{LM}, \alpha_{LK}, \alpha_{MK}, \beta_{exp}^D)$, and estimate these parameters by GMM with moment conditions

$$E[Z'_{jt} \epsilon_{jt}^\omega(\alpha_L, \alpha_M, \alpha_K, \alpha_{LL}, \alpha_{MM}, \alpha_{KK}, \alpha_{LM}, \alpha_{LK}, \alpha_{MK}, \beta_{exp}^D)] = 0, \quad (19)$$

where $Z_{jt} = (l_{jt-1}, m_{jt-1}, k_{jt}, l_{jt-1}^2, m_{jt-1}^2, k_{jt}^2, l_{jt-1} m_{jt-1}, l_{jt-1} k_{jt}, m_{jt-1} k_{jt}, D_{jt-1}^{exp})$ is the instrument vector. We use the estimation results of the translog-form production function as a robustness check and report the estimates of the Cobb-Douglas production function as the main discussion.

4 Estimation Results: Productivity, Markup, and Returns to Scale

We apply the model to the Chinese manufacturing industry from 2000 to 2006. This section first reports the estimation results. Then we calculate the production efficiency gains from export and the implications on firm profitability and consumer prices.

4.1 Returns to Scale and Export Premium on Productivity and Demand

Table 3 reports the estimation results. Our benchmark and IV results generate similar patterns. There are three main findings. First, export has an economically and statistically significant effect on productivity, as captured by the parameter β_{exp}^ω . This echoes the findings in Li et al. (2025) in their researched industries (leather shoes, shirts, and suits). Exporting raises productivity by about 0.8% on average in both the benchmark and IV specifications. This result is at the higher end of the estimates, compared with the literature that uses revenue data without correcting for the unobserved output price bias (e.g. Lileeva and Trefler, 2010; Grieco et al., 2022). The improved productivity reduces the exporters' production costs, giving them competitive advantages in the market and potentially a higher profit.

Second, exporting increases the firm’s market size. The demand-parameter estimate is $\hat{\beta}_{exp}^D = 0.034$ (SE = 0.0027) in the IV specification and 0.019 (SE = 0.0016) in the benchmark specification. Under the demand system in (6) the firm-level demand effect is heterogeneous across firms and is given by $\beta_{exp}^D/(\mu - 1)$; its sample mean is 15.4% in the IV specification and 8.2% in the benchmark (Table 3, Panel B). The increased demand shifter gives the exporter a larger market size, potentially leading to higher profitability for the firm.

Third, firm-level production demonstrates substantial increasing returns to scale after correcting for firm heterogeneity in markup. In the table, the (total) returns to scale are captured by $S(RTS)$, which is the sum of output elasticities for capital, labor, and material. In our main estimation, total returns to scale (RTS) = 1.089 (IV) and 1.103 (Benchmark), with variable-input returns (RTSV) = 1.055 and 1.069, respectively. This result is consistent with findings in [De Loecker et al. \(2016a\)](#), who use the output quantity data to estimate the translog production function with the adjustment for the potential input price bias. They find that 68% of the sample exhibit increasing returns to scale. Moreover, they find that firms’ marginal cost decreases in output quantity. As shown later, this result is robust using different specifications of production functions and other estimation methods. The estimate is higher than that reported in the literature using revenue data, in which the unobserved firm heterogeneity in markup biases down the estimate of returns to scale, as discussed in Section 3.

Increasing returns to scale provides a new source of gains from exports. Given that firms have a larger market and produce more after export, increasing returns to scale reduces the firm’s marginal costs and improves its efficiency. The result echoes early studies [Scherer \(1980\)](#) and [Panzar \(1989\)](#), who illustrated the existence of economies of scale in both case studies and theory. This result also provides empirical support to the new growth theory ([Romer, 1986](#)) and new trade theory ([Krugman, 1980a, 1995](#)), whose results are mainly based on the assumption of increasing returns to scale at the aggregate economy. Our results show that increasing returns to scale exist even at the firm level.

As a comparison, columns (3)-(4) report the estimation results using the widely-used approaches in the literature based on revenue data to estimate the revenue production function (8) without correcting for firm heterogeneity in markup. Column (3) uses the original revenue as a proxy for output, and column (4) uses the deflated revenue instead. As shown in the table, without correcting for firm heterogeneity in prices (and markup), the traditional methods find a negligible effect of export on revenue productivity (0.2-0.3%), which is one order lower in magnitude compared with our results. This is in contrast to [Garcia-Marin and Voigtländer \(2019\)](#), which shows that TFPR would increase as long as markup increases (to be shown

Table 3: Estimation Results

Parameter	(1)	(2)	(3)	(4)	(5)
	(IV)	Ours (Benchmark)	K&G(1996)	Original	Deflated
Panel A: Estimates of Production Function					
β_{exp}^ω (Productivity Effect)	0.008 (0.0008)	0.008 (0.0005)	0.002 (0.0008)	0.002 (0.0002)	0.003 (0.0003)
β_{exp}^D (Demand Parameter)	0.034 (0.0027)	0.019 (0.0016)	0.024 (0.0054)	0.015 (0.0009)	0.011 (0.0011)
RTS	1.089 (0.0068)	1.103 (0.0053)	1.007 (0.0087)	0.968 (0.0027)	0.981 (0.0041)
RTSV	1.055 (0.0074)	1.069 (0.0057)	0.985 (0.0193)	0.954 (0.0040)	0.965 (0.0059)
α_L	0.080 (0.0009)	0.084 (0.0009)	0.058 (0.0015)	0.041 (0.0004)	0.042 (0.0004)
α_M	0.974 (0.0067)	0.985 (0.0052)	0.927 (0.0207)	0.913 (0.0041)	0.923 (0.0059)
α_K	0.034 (0.0008)	0.033 (0.0006)	0.023 (0.0106)	0.014 (0.0014)	0.016 (0.0018)
Panel B: The Implied Demand Effect of Export					
Mean	0.154	0.082	0.024	0.015	0.011
Median	0.131	0.071			
SD	0.086	0.043			
IQR	0.109	0.056			
IDR	0.220	0.111			
Observations	787,247	1,234,292	1,234,292	1,234,292	1,234,292

Note 1: Columns (1)–(2) implement our proposed method (Section 3.2). Column (1) instruments the export dummy D_{jt}^{exp} with the interaction of the U.S.-imposed industry-level Normal Trade Relations (NTR) tariff gap, the post-WTO-accession indicator, and the firm’s lagged export status; Column (2) is the same specification without instrumentation. The IV sample is smaller because of the lagged-export-status requirement.

Note 2: Columns (3)–(5) implement alternative procedures from the literature. Column (3) follows Klette and Griliches (1996), imposing a constant industry-level markup to partially absorb the market-power bias. Column (4) estimates the revenue production function directly. Column (5) deflates revenue with the industry-level price index of Brandt et al. (2017) before estimation.

Note 3: Panel B reports the export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ implied by the demand system in equation (6), which is heterogeneous across firms with differing market power. For Columns (1)–(2), we report the mean, median, standard deviation, IQR, and IDR of this firm-level effect after symmetric 10% truncation at both tails. For Columns (3)–(5), the demand parameter enters the revenue production function as a constant elasticity, so Panel B reports its point estimate in the mean row only.

Note 4: Standard errors clustered at the firm level are reported in parentheses.

in Section 4.2) under constant returns to scale. Our results are intuitive. As discussed in Section 2.3, with increasing returns to scale, TFPR may not increase even if export increases markup because increasing returns to scale with enlarged market size drives down TFPR.¹³

The revenue-based approaches in the literature also underestimate the effect of export on market size, as shown in the estimates of β_{exp}^D in columns (4) and (5). Notice that the markup-adjusted revenue function in (11) can be rewritten as: $r_{jt} - \bar{r}_{Jt} = \frac{1}{\mu_{jt}}(\alpha_L l_{jt} + \alpha_M m_{jt} + \alpha_K k_{jt} + \omega_{jt} - \bar{y}_{Jt} + \beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D)$. The term $(1/\mu_{jt})$ before the export variable D_{jt}^{exp} biases down the true estimates of export effects on productivity. Moreover, by ignoring firm heterogeneity in markup, the literature also underestimates the returns to scale as shown in columns (4) and (5). This is as expected following our discussion in Equation (8).

Interestingly, when controlling for a common markup at the industry level when estimating the revenue production function (8) using the method proposed by Klette and Griliches (1996) (Column (3) of Table 3), the estimated returns to scale rise to 1.007, markedly larger than the 0.968 and 0.981 obtained in Columns (4)–(5) under the traditional approaches. This is consistent with the insights and results in Klette and Griliches (1996). However, it is still lower than our estimate because it ignores firm heterogeneity in markup. More importantly, the Klette-Griliches approach still substantially understates the productivity and demand gains from exports. This is because, by assuming constant demand elasticity, the approach fails to capture the changes in markup before and after the firm starts exporting, which has an impact on the estimation of the productivity and demand premium of exports.

Firm Heterogeneity in Markup and Productivity

After estimating the output elasticities, we can calculate the firm-level markup using (10) and productivity using (13). We plot their kernel density distributions in Figure 2.

The markup is greater than 1 on average, with the mean at 1.225 and median at 1.247. This explains why estimating the revenue production function without correcting for markup may understate the returns to scale. There is also substantial heterogeneity in markup across firms/years, with a cross-sectional standard deviation of 0.252; the markup difference between the 90th and 10th percentiles is roughly 0.4 log points.

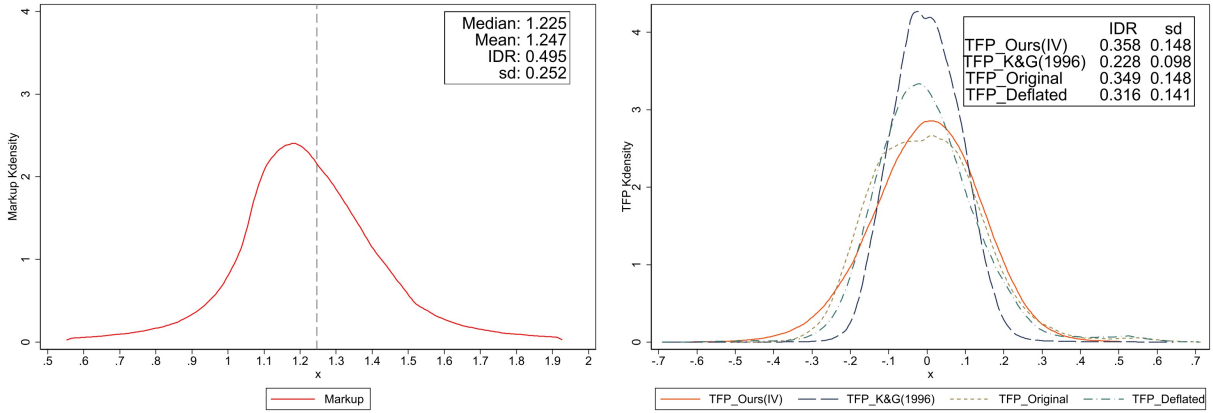
The heterogeneity in markup exceeds that in productivity. The cross-sectional dispersion of markups (SD = 0.252) is larger than that of TFP itself (SD = 0.148), so markup heterogeneity

¹³Using the updated estimates of $\Delta \ln \mu^{exp}$, β_{exp}^ω , and Δr^{exp} in (3), the implied change in TFPR after export remains close to zero, consistent with the near-zero TFPR effect of export in Columns (3)–(5) of Table 3. The decomposition is reported in Appendix Table B1.

— not productivity heterogeneity — drives most of the variation in revenue-based TFP.

This result highlights the importance of taking into account firm heterogeneity in markup in the analysis of firm heterogeneity and performance. This result also stresses why controlling common markup as in [Klette and Griliches \(1996\)](#) may still substantially bias the estimates of productivity and the export premium. Meanwhile, the different dispersions of productivity estimated by different methods also suggest different heterogeneity levels in firms. Our method suggests that firms are more heterogeneous in terms of productivity than the traditional estimates.

Figure 2: Markup and TFP Dispersion (LP)



Note: The left panel displays the distribution of estimated firm-level markups (μ_{jt}); the mean and median are 1.225 and 1.247, respectively. The right panel displays the distribution of estimated TFP (ω_{jt}) from our method and from three traditional methods, each normalized by its sample mean. The cross-sectional dispersion of markups ($SD = 0.252$) exceeds that of TFP itself ($SD = 0.148$), reinforcing the concern formalized in equation (3) that markup heterogeneity, rather than productivity heterogeneity, drives most of the variation in revenue-based TFP (TFPR).

4.2 Export Premium on Markup

The larger market size and improved efficiency may allow the exporters to charge a higher markup. In this section, we estimate the impact of export on markup based on the following equation:

$$\mu_{jt} = \beta_0^\mu + \beta_{exp}^\mu D_{jt}^{exp} + \beta^X \mathbf{X}_{jt} + \gamma_j + \gamma_t + \xi_{jt}^\mu, \quad (20)$$

where D_{jt}^{exp} is the export dummy as in equation (5) and our parameter of interest, β_{exp}^μ , captures the average exporting effect on markup. We control for a set of firm characteristics

$\mathbf{X}_{jt}$, including firm size (measured by labor employed) and capital intensity. We also control for firm FE and time FE to capture time-invariant firm characteristics and common time shocks. ξ_{jt}^μ is an i.i.d shock to the firm's markup. The estimation results are reported in Table 4.

Table 4: Exporting Effect on Markup (LP)

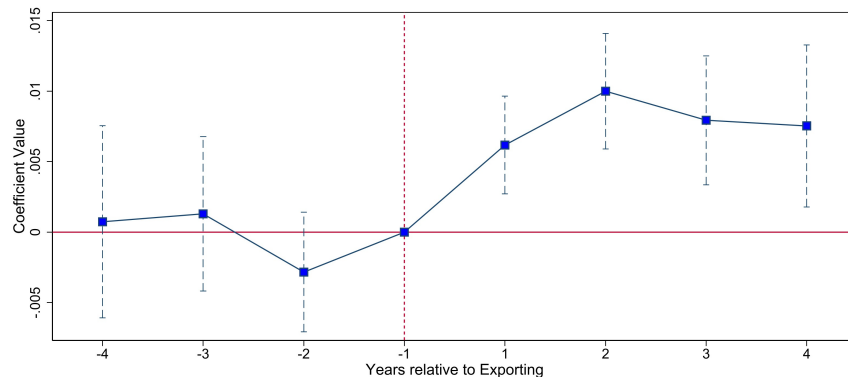
	(1)	(2)
	Markup	Markup
D_{jt}^{exp}	0.008*** (0.0015)	0.010*** (0.0015)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.326	0.326

Note: Standard errors (clustered at the firm level) in parentheses.

* $p < .10$, ** $p < .05$, *** $p < .01$

Column (1) reports the estimation results after controlling the two fixed effects, and column (2) further controls for the firm-specific variables, including firm size and capital intensity. We find that export increases a firm's markup by 0.8-1.0 percentage points. This result suggests that exporters are able to charge a higher markup after starting export, presumably because of the increased market size and improved production efficiency.

Figure 3: Dynamic Exporting Effect on Markup (LP)



Note: The range represents 95% confidence interval of the parameter estimates.

We also estimate the pre-trend and dynamic effects of export on markup, based on an extended model of (20). The results are reported in Figure 3.¹⁴ On the horizontal axis of the figure,

¹⁴We only kept four years before and four years after exporting in the dynamic graph because our database

“-1” represents the year right before export, and “1” represents the first year of exporting. The other years are similarly defined. The result shows no obvious pre-trend. Before they export, exporters’ markups are not significantly different from those of non-exporters. It also confirms the large dynamic effect of export on markup. Compared with non-exporters, exporters’ markup increases significantly from the first year of export, and the effects remain multiple years after exporting.

5 Validation with Output Quantity

When output quantity data are not available, the unobserved markup causes a bias in productivity and returns to scale when estimated using the revenue production function. When output quantity data are available, however, the quantity-based production function should generate consistent estimation results. We validate our new method by comparing our estimation results with that based on quantity-based output production function using a smaller sample from 2000 to 2006, which has output quantity at the firm level. The output quantity data come from the output quantity survey as discussed in Section 2.1 above. After merging it with our main data, there are about 444,475 observations with output quantity.

We restrict the sample to single-product firms for three complementary reasons. First, the data report quantity at the firm-product level but revenue at the firm level, so for multi-product firms, we cannot construct an accurate firm-level output-price index without further assumptions about the within-firm product mix. Restricting to single-product firms makes the firm-level output price cleanly identified as the ratio of total output revenue to reported output quantity. Second, all production-function inputs (labor, materials, capital) are reported at the firm level, whereas a multi-product firm uses these inputs jointly to manufacture several distinct products. Without observing how inputs are allocated across products, the estimated output elasticities for multi-product firms would conflate the potentially different input-output relationships of several different production technologies. Restricting to single-product firms shuts this input-allocation problem down by construction. Third, in multi-product firms, it is difficult to separate returns to scale from economies of scope without additional assumptions and data. Restricting to single-product firms therefore cleanly identifies returns to scale within a single product line, free of any contamination from scope economies. After this restriction, the sample contains 321,280 observations (118,671 firms) with output-quantity

only covers the years from 2000 to 2006. Because we do not know when a firm started exporting if it was already exporting in 2000, the maximum number of years after starting to export that we can identify is 6 years (for those firms that started exporting in 2001). However, the number of observations that were 5 or 6 years before or after the firm started exporting is small, making the estimation of the dynamic effects at 5 and 6 years inaccurate. Therefore, we do not include them in the analysis of dynamic effects.

data, of which 19,838 firms (16.72%) are exporters.

One caveat of estimating the physical output production function, as [De Loecker et al. \(2016a\)](#) and [Li and Zhang \(2022\)](#) point out, is that output quality and input quality (and output prices and input prices) may be positively correlated. This correlation may bias our estimates. This problem is especially serious when the physical output quantity is used as output, but the (deflated) input expenditure is used as a proxy for input quantity, as in most applications, including ours. In this case, the input quality and price are partially controlled by the input expenditure, but output quality enters the error term, which biases downward the estimates of output elasticities and thus returns to scale. In fact, even if we don't consider quality differences, input prices and output prices are still positively correlated due to the optimal pricing strategy of firms facing imperfect competition, leading to a similar bias in the production function estimation.

To address this potential problem, we follow the insight of [De Loecker et al. \(2016a\)](#) to use the output price to control for input price differences. Specifically, we assume that the firm-level output prices are determined by variations in firm-level input prices and the market-level input and output price indices, so we can invert the output prices to construct an index of firm-level input prices using variations of firm-level output prices and market-level input and output price indices. Then, the “physical” input quantity is calculated by dividing the input expenditure by the firm-level input price index. We use it in the production function estimation. The data construction details, together with the detailed estimation process, are discussed in Appendix [A.9](#).

Table [5](#) reports the estimation results based on the observed output quantity. As a comparison, we also report the estimation results using our method based on revenue data in this small sample. There are three major findings. First, the physical production function shows significant increasing returns to scale, at 1.069 in the IV specification and 1.079 in the benchmark specification, as reported in Columns (1) and (2) in Table [5](#). The result is consistent with [De Loecker et al. \(2016a\)](#), who find that 68% of their sample observations show increasing returns to scale by estimating the physical quantity production function after adjusting for the potential biases caused by the input prices/quality. Our result is also consistent with [Lu and Yu \(2015\)](#), who use the same output quantity data and find the average RTS of Chinese manufacturing firms is greater than 1 ($RTS = 1.125$) after addressing the input price biases.

Second, exports significantly raise firm productivity, showing an export premium of 0.4–0.5% in the IV and Benchmark specifications in Columns (1)–(2). The effect is slightly smaller

Table 5: Results Using Quantity Data, Our Approach, and the Revenue-Based Methods

	(1)	(2)	(3)	(4)	(5)	(6)
	Quantity Output		Revenue Output			
			Ours		K&G(1996)	Deflated
Parameter	(IV)	(Benchmark)	(IV)	(Benchmark)		
Panel A: Estimates of Production Function						
β_{exp}^ω (Productivity Effect)	0.004 (0.0009)	0.005 (0.0006)	0.005 (0.0008)	0.006 (0.0023)	0.002 (0.0008)	0.001 (0.0003)
β_{exp}^D (Demand Parameter)	0.035 (0.0202)	0.101 (0.0134)	0.012 (0.0027)	0.019 (0.0043)	0.029 (0.0046)	0.025 (0.0017)
RTS	1.069 (0.0003)	1.079 (0.0190)	1.038 (0.0020)	1.040 (0.0228)	1.038 (0.0039)	0.998 (0.0009)
RTSV	1.020 (0.0003)	1.021 (0.0008)	1.010 (0.0016)	1.018 (0.0216)	1.023 (0.0009)	0.997 (0.0007)
α_L	0.019 (0.0008)	0.019 (0.0006)	0.053 (0.0012)	0.055 (0.0013)	0.038 (0.0010)	0.023 (0.0006)
α_M	1.001 (0.0012)	1.003 (0.0005)	0.957 (0.0015)	0.964 (0.0207)	0.984 (0.0012)	0.974 (0.0004)
α_K	0.048 (0.0033)	0.057 (0.0188)	0.028 (0.0014)	0.021 (0.0014)	0.015 (0.0043)	0.001 (0.0008)
Panel B: The Implied Demand Effect of Export						
Mean	0.184	0.519	0.126	0.065	0.029	0.025
Median	0.153	0.433	0.105	0.054		
SD	0.118	0.350	0.096	0.045		
IQR	0.141	0.402	0.103	0.051		
IDR	0.287	0.833	0.217	0.106		
Observations	182,371	321,282	182,371	321,282	321,280	321,280

Note 1: Columns (1)–(2) report estimates from a quantity-based production function on the single-product sample (2000–2006, Section 5). Following De Loecker et al. (2016a), the firm-level input quantity is constructed by deflating input expenditure with a firm-level input price index inverted from output prices; details are in Appendix A.9. Column (1) instruments D_{jt}^{exp} as in Table 3; Column (2) is the same specification without instrumentation.

Note 2: Columns (3)–(6) report revenue-based estimates on the same single-product sample. Columns (3)–(4) apply our proposed method, with IV and benchmark specifications defined as in Table 3. Column (5) follows Klette and Griliches (1996), imposing a constant industry-level markup. Column (6) deflates revenue with the industry-level price index of Brandt et al. (2017). All revenue-based specifications use the Levinsohn–Petrin first stage.

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across firms with differing market power. For Columns (1)–(2), β_{exp}^D is identified sequentially from the quantity-based PF: combining α_M with the materials FOC yields the De Loecker and Warzynski (2012) markup $\mu_{jt} = \alpha_M/s_{jt}^M$, and inverting equation (6) gives the closed-form residual $z_{jt} = \mu_{jt}(p_{jt} - \bar{p}_{Jt}) + (\mu_{jt} - 1)(y_{jt} - \bar{y}_{Jt})$, which we regress on D_{jt}^{exp} . For Columns (3)–(4), β_{exp}^D is identified jointly with the production function in the revenue specification. The mean, median, SD, IQR, and IDR for Columns (1)–(4) are computed after symmetric 10% truncation at both tails. Columns (5)–(6) impose a constant demand elasticity, so Panel B reports only the point estimate (mean row).

Note 4: Standard errors clustered at the firm level are reported in parentheses.

than that in our main results using the full sample, but the difference may be driven by the sample differences. When estimating the model using our revenue-based method on the same single-product sample (pretending output-quantity data are unavailable), we obtain very similar results: export has a productivity premium of 0.5–0.6% and total returns to scale is 1.038–1.040 as reported in Columns (3) and (4). The demand parameter estimate is $\hat{\beta}_{exp}^D = 0.019$ (SE = 0.0043) in the IV specification and 0.012 (SE = 0.0027) in the benchmark, implying a mean export demand premium of 12.6% (IV) and 6.5% (Benchmark). These patterns are very close to those we found using our method in the benchmark model.

Third, consistent with our main results, the traditional revenue-based estimation approaches underestimate both the productivity effect (0.1%) and the returns to scale (0.998) even after deflating revenue by an industry-year-level output deflator, as reported in Column (6) of Table 5. Imposing a constant markup as in Klette and Griliches (1996) (Column (5)) detects modest IRS (1.038) but still misses the productivity premium. The results further show that our results and the new estimation methods are robust.

6 Gains from Exporting

Improved productivity and increasing returns to scale imply that export can improve production efficiency. This efficiency, together with increased markup, guarantees that exporters can gain a greater profit. Moreover, can consumers gain (or lose) from exports by paying lower (or higher) output prices? This section answers these questions. First, we calculate the changes in production efficiency contributed by improved productivity and increasing returns to scale. Then, we calculate the impact on firms and consumers based on changes in their profitability and paid prices, respectively.

6.1 Export and Production Efficiency

To quantify the export premium in production efficiency and evaluate the contribution of improved productivity and increasing returns to scale, first note that the cost minimization problem as discussed in Section 3 implies the following marginal costs function¹⁵,

$$mc_{jt} = \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} y_{jt} - \frac{1}{\alpha_L + \alpha_M} \omega_{jt} + \frac{\alpha_M}{\alpha_L + \alpha_M} p_{jt}^M + \frac{\alpha_L}{\alpha_L + \alpha_M} p_{jt}^L - \frac{\alpha_M}{\alpha_L + \alpha_M} \ln(\alpha_M) - \frac{\alpha_L}{\alpha_L + \alpha_M} \ln(\alpha_L) - \frac{\alpha_K}{\alpha_L + \alpha_M} k_{jt} - \frac{1}{\alpha_L + \alpha_M} \xi_{jt}, \quad (21)$$

¹⁵Detailed derivation process is reported in Appendix A.7.

where mc_{jt} is the log form of marginal cost, and other variables are similarly defined as before. We ignore the indirect effect of exports on input prices and capital stock.¹⁶ Hence, the impact of export on marginal costs can be written in the following form,

$$\Delta mc^{exp} = \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} \Delta y^{exp} - \frac{1}{\alpha_L + \alpha_M} \beta_{exp}^\omega, \quad (22)$$

where Δy^{exp} denotes the increase in output caused by exporting. The first term on the right, $\frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} \Delta y_j^{exp}$, captures the exporter's efficiency gains from increasing returns to scale. Given increasing returns to scale of variable inputs ($\alpha_L + \alpha_M > 1$), producing more after export reduces the firm's marginal costs at a rate of $\frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M}$. The second term, $\frac{1}{\alpha_L + \alpha_M} \beta_{exp}^\omega$, captures the efficiency gains through improved productivity. Because $\beta_{exp}^\omega > 0$, export reduces marginal costs.¹⁷

In (22), changes in output quantities Δy^{exp} are not observed in the data. It can be derived using (22), combined with the definition of (changes in) markup and revenue, as follows

$$\Delta \ln \mu^{exp} = \Delta p^{exp} - \Delta mc^{exp}, \quad (23)$$

$$\Delta r^{exp} = \Delta p^{exp} + \Delta y^{exp}. \quad (24)$$

where Δx represents changes in variable x as usual. The average changes in (logarithm) markup after export, $\Delta \ln \mu^{exp}$, has been estimated in Section 4.2 at 1%.¹⁸ The average changes in revenue after export, Δr^{exp} , can also be estimated similarly, which is 20.974%. Combining (22)-(24) yields the solution for Δp^{exp} , Δy^{exp} and Δmc^{exp} as the functions of observed estimates ($\Delta \ln \mu^{exp}$, Δr^{exp} , β_{exp}^ω) as follows:

$$\begin{aligned} \Delta y^{exp} &= (\alpha_L + \alpha_M) [\Delta r^{exp} - \Delta \ln(\mu)^{exp}] + \beta_{exp}^\omega \\ \Delta p^{exp} &= [1 - (\alpha_L + \alpha_M)] \Delta r^{exp} + (\alpha_L + \alpha_M) \Delta \ln(\mu)^{exp} - \beta_{exp}^\omega \\ \Delta mc^{exp} &= [1 - (\alpha_L + \alpha_M)] [\Delta r^{exp} - \Delta \ln(\mu)^{exp}] - \beta_{exp}^\omega \end{aligned} \quad (25)$$

Taking (25) into the data, we calculate the average exporting changes in output quantities Δy^{exp} , output prices Δp^{exp} , and marginal costs Δmc^{exp} , which equal approximately 21.8%,

¹⁶By focusing on the direct gains from export via improved productivity and increasing returns to scale and ignoring the indirect gains from adjusting capital investment and input prices, our result can be thought of as the lower bound of the total gains from export.

¹⁷Of course, because improved productivity also induces more output, further allowing the exporter to gain more from increasing returns to scale (captured by the first term). Hence, more precisely, the second term captures the direct effect of export on production costs via improved productivity.

¹⁸Export's effect on log markup almost equals the export's effect on markup numerically. See Table B6 for details.

−0.854%, and −1.854%, respectively.

With changes in output quantity in hand, we can calculate the contribution of increasing returns to scale to production efficiency, as well as the impact of export on production efficiency based on (22). The results are summarized in Table 6. Exporting reduces exporters’ marginal cost of production by 1.854% on average, materially improving production efficiency. Of this decline, 0.720 percentage points (38.703%) come through the productivity-premium channel, while the remaining 1.134 percentage points (61.297%) operate through increasing returns to scale. The IRS channel thus accounts for nearly two thirds of the production-efficiency gain from exporting — a magnitude that the conventional revenue-based methods miss entirely.

Table 6: Gains from Export Decomposition

Marginal Cost ↓ 1.854%		Markup	Consumer Price	Exporter Profits
<i>through IRS</i>	<i>through Productivity</i>			
↓ 1.134%	↓ 0.720%	↑ 1%	↓ 0.854%	↑ 25.384%

Note 1: This table decomposes the gains from exporting. According to equation (22), exporters’ marginal cost declines by 1.854% on average within the first year of exporting. Of this decline, 61.297% (1.134 percentage points) operates through increasing returns to scale (IRS), and the remaining 38.703% (0.720 percentage points) is attributable to the export productivity premium.

Note 2: As reported in Table B6, the markup rises by 1% on average upon exporting. Combining the markup increase with the marginal-cost decline through the decomposition in equation (23) implies that the consumer price falls by 0.854% on average.

Note 3: Applying the decomposition in equation (26), exporter profits increase by 25.384% on average.

6.2 Benefits to Firms and Consumers

Firm profitability. The increased production efficiency and markup naturally imply a gain in profitability for exporters. To quantify this effect, we calculate the changes in profits as implied by the model estimates. By the definition of profit $\Pi_{jt} = R_{jt} - (P_{jt}^L L_{jt} + P_{jt}^M M_{jt})$ and the definition of raw markup $\tilde{\mu}_{jt} \equiv R_{jt} / (P_{jt}^L L_{jt} + P_{jt}^M M_{jt})$, we can express the profit as $\Pi_{jt} = R_{jt}(\tilde{\mu}_{jt} - 1) / \tilde{\mu}_{jt}$. As shown in Appendix A.8, the first-order approximation of the impact of export on firms’ profits can be written in the following form:

$$\Delta\pi^{exp} \simeq \Delta y^{exp} + \Delta p^{exp} + \frac{1}{\tilde{\mu}^{exp=0}(\tilde{\mu}^{exp=0} - 1)} \Delta\tilde{\mu}^{exp} \quad (26)$$

The changes in output quantity, prices, and markup after exporting have been calculated above. We further use the raw markup of all non-exporting observations to represent $\tilde{\mu}^{exp=0}$. Combining all information, we find that exporting raises firms’ profit by 25.384% on average (Table 6). A large part of this is due to the increased market size, but the improved efficiency and markup further improve the exporter’s profits.

Consumer welfare. Given the increased markup after export, can consumers benefit from the improved efficiency? Since the markup in the log form can be expressed as $\ln(\mu_{jt}) = p_{jt} - mc_{jt}$, we can derive the changes in consumer prices from information on changes in markup and marginal costs. The calculation above implies that the efficiency effect (-1.854%) dominates the markup increase ($+1\%$), so the average price paid by consumers for products sold by firms that start exporting falls by 0.854% (Table 6). This means that although firms increase markup after exporting, they are able to pass through some of the export’s efficiency premium to consumers by lowering the output prices.

Note that the output price of an exporter here is the firm-level average price, including both export and domestic sales. As a result, the reduction in exporters’ output price may result from a lower export price or a reduced price for domestic consumers. In other words, it is possible that both foreign buyers and domestic consumers can benefit from improved efficiency, although more information is needed to answer this question precisely.

The back-of-the-envelope calculation in this subsection shows that both firms and consumers benefit from exports. After exporting, firms’ profits increase by about a quarter due to increased productivity, markup, increasing returns to scale, and larger market size. Yet, due to improved production efficiency, firms are able to charge a lower price to achieve higher markup and the lower prices benefit consumer welfare.

7 Robustness Checks

This section shows that our results are robust to an alternative estimation approach and a more flexible production function that allows for variable output elasticities.

7.1 Single Product Firms

We used the full sample, inclusive of the single-product and multi-product firms, in our main estimation. One potential concern is that the documented increasing returns to scale and the export productivity premium could be driven by economies of scope across product lines in multi-product firms rather than by genuine economies of scale within a single product. To address this concern, we re-estimate the model on a restricted sample of single-product firms—defined as firms that report only one product in the main-product field of the ASIE survey—so that any within-firm reallocation across products is shut down by construction.

The estimation results, reported in Table 7, are consistent with the full-sample results in Table 3. Under our method, exporting raises productivity by about $0.4\text{--}0.6\%$ in the IV

Table 7: Estimation Results (Single-Product Firms)

Parameter	(1)	(2)	(3)	(4)	(5)
		Ours	K&G(1996)	Original	Deflated
	(IV)	(Benchmark)			
Panel A: Estimates of Production Function					
β_{exp}^ω (Productivity Effect)	0.004 (0.0010)	0.006 (0.0006)	0.002 (0.0008)	0.001 (0.0003)	0.003 (0.0003)
β_{exp}^D (Demand Parameter)	0.039 (0.0032)	0.021 (0.0016)	0.026 (0.0086)	0.015 (0.0010)	0.011 (0.0011)
RTS	1.068 (0.0161)	1.082 (0.0088)	1.014 (0.0096)	0.970 (0.0020)	0.979 (0.0025)
RTSV	1.033 (0.0170)	1.047 (0.0095)	0.989 (0.0243)	0.953 (0.0028)	0.962 (0.0035)
α_L	0.077 (0.0016)	0.080 (0.0010)	0.057 (0.0022)	0.041 (0.0005)	0.041 (0.0005)
α_M	0.956 (0.0156)	0.967 (0.0087)	0.932 (0.0264)	0.913 (0.0028)	0.920 (0.0035)
α_K	0.036 (0.0012)	0.036 (0.0009)	0.025 (0.0148)	0.016 (0.0012)	0.017 (0.0012)
Panel B: The Implied Demand Effect of Export					
Mean	0.190	0.095	0.026	0.015	0.011
Median	0.159	0.081			
SD	0.118	0.054			
IQR	0.142	0.068			
IDR	0.293	0.138			
Observations	535,168	915,512	915,512	915,512	915,512

Note 1: This table re-estimates Table 3 on the subsample of single-product firms (those reporting a single product in the main-product field of the ASIE survey), to rule out within-firm reallocation across products as a source of the documented returns-to-scale and export-premium estimates. Column conventions and the IV strategy follow Table 3.

Note 2: Columns (3)–(5) implement alternative procedures from the literature on the same single-product sample. Column (3) follows Klette and Griliches (1996), imposing a constant industry-level markup. Column (4) estimates the revenue production function directly. Column (5) deflates revenue with the industry-level price index of Brandt et al. (2017).

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across firms with differing market power. For Columns (1)–(2), we report the mean, median, SD, IQR, and IDR of this firm-level effect after symmetric 10% truncation at both tails. For Columns (3)–(5), the demand parameter enters the revenue production function as a constant elasticity, so Panel B reports its point estimate in the mean row only.

Note 4: Standard errors clustered at the firm level are reported in parentheses.

and benchmark specifications. Production displays significant increasing returns to scale, at 1.068 and 1.082 for total returns to scale for these two specifications and 1.033 / 1.047 for variable returns to scale. The traditional revenue-based methods (Columns (3)–(5)) again substantially understate both margins: the productivity premium is essentially zero, and returns to scale are at or slightly below one. As in the full sample, the export-productivity premium estimated by our method is about two to three times that detected by the traditional methods, and the rank ordering of methods in terms of RTS is unchanged.

The demand-side patterns also mirror the main results. The demand parameter (β_{exp}^D) is estimated at 0.039 (SE = 0.0032) in the IV specification and 0.021 (SE = 0.0016) in the benchmark, both slightly larger than the corresponding full-sample estimates of 0.034 and 0.019 in Table 3. The implied mean firm-level demand effects of 19.0% (IV) and 9.5% (Benchmark) are close to the estimates in the full sample (15.4% and 8.2%).

Taken together, these results indicate that our main conclusions—a significant productivity premium of exporting, substantial increasing returns to scale, and a positive demand effect of foreign-market access—are not driven by within-firm reallocation across products in multi-product firms. The documented increasing returns to scale therefore reflect genuine economies of scale within a product line, rather than artifacts of economies of scope.

7.2 ACF Estimation

In our main results, we used [Levinsohn and Petrin \(2003\)](#). In this section, we use the alternative method provided by [Akerberg et al. \(2015\)](#) to estimate all parameters jointly. The structural estimation results of all the specifications are displayed in Table 8.

The first two rows of Table 8 show that the exporting effects on productivity and demand shifter estimated using ACF are very similar to our main results as reported in Table 3. Besides, the exporting effects on productivity detected by the traditional methods are very small, as expected. The exporting effect on productivity using our method is still about five times as large as the three traditional methods, indicating that our discussion in the previous sections is robust. In terms of the output elasticity, notice that our method and [Klette and Griliches \(1996\)](#)’s method predict increasing returns to scale, while the original revenue method and deflated revenue method show decreasing returns to scale. This, again, is consistent with the main results using [Levinsohn and Petrin \(2003\)](#).

After estimating the production function, we calculated the new markup. We also conducted a similar analysis of export’s effect on markup as in Section 4.2. As shown in Table B2, the impact of export on markup based on ACF is almost the same as that in LP’s case.

Table 8: Estimation Results (ACF)

Parameter	(1)	(2)	(3)	(4)	(5)
	(IV)	Ours (Benchmark)	K&G(1996)	Original	Deflated
Panel A: Estimates of Production Function					
β_{exp}^ω (Productivity Effect)	0.007 (0.0010)	0.008 (0.0005)	-0.003 (0.0128)	0.001 (0.0003)	0.002 (0.0008)
β_{exp}^D (Demand Parameter)	0.038 (0.0033)	0.025 (0.0016)	0.032 (0.0400)	0.019 (0.0011)	0.016 (0.0024)
RTS	1.097 (0.0067)	1.100 (0.0056)	1.054 (0.0316)	0.964 (0.0049)	0.970 (0.0170)
RTSV	1.064 (0.0072)	1.068 (0.0058)	1.041 (0.0311)	0.943 (0.0072)	0.945 (0.0248)
α_L	0.064 (0.0033)	0.047 (0.0042)	0.122 (0.0816)	0.055 (0.0026)	0.061 (0.0040)
α_M	1.000 (0.0053)	1.021 (0.0029)	0.919 (0.0635)	0.888 (0.0090)	0.884 (0.0275)
α_K	0.033 (0.0007)	0.033 (0.0005)	0.013 (0.0143)	0.021 (0.0024)	0.025 (0.0078)
Panel B: The Implied Demand Effect of Export					
Mean	0.164	0.104	0.032	0.019	0.016
Median	0.141	0.090			
SD	0.089	0.056			
IQR	0.113	0.071			
IDR	0.227	0.143			
Observations	787,247	1,234,292	1,234,292	1,234,292	1,234,292

Note 1: This table replicates Table 3 as a robustness check, replacing the Levinsohn–Petrin first stage with the [Akerberg et al. \(2015\)](#) (ACF) procedure throughout. Columns (1)–(2) implement our proposed method; column conventions and the IV strategy follow Table 3.

Note 2: Columns (3)–(5) implement alternative procedures from the literature, each with the ACF first stage where applicable. Column (3) follows [Klette and Griliches \(1996\)](#), imposing a constant industry-level markup. Column (4) estimates the revenue production function directly. Column (5) deflates revenue with the industry-level price index of [Brandt et al. \(2017\)](#).

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across firms with differing market power. For Columns (1)–(2), we report the mean, median, SD, IQR, and IDR of this firm-level effect after symmetric 10% truncation at both tails. For Columns (3)–(5), the demand parameter enters the revenue production function as a constant elasticity, so Panel B reports its point estimate in the mean row only.

Note 4: Standard errors clustered at the firm level are reported in parentheses.

As in the main result, the increased productivity after export and increasing returns to scale (together with a larger market) imply an export premium on production efficiency. Under ACF, exporting reduces exporters’ marginal cost by 1.711% on average, of which 1.098 percentage points (64.173%) operate through increasing returns to scale and the remaining 0.613 percentage points (35.827%) through the productivity-premium channel (Table B3). The improved production efficiency and the higher markup¹⁹ jointly raise exporter profits by 25.346% on average and reduce consumer prices by 0.711% (Table B3). These magnitudes are quantitatively close to the LP-based main results in Table 6 (25.384% and 0.854%, respectively), confirming that our main conclusions are not sensitive to the choice of first-stage productivity estimator.

7.3 Flexibility of Variable Inputs

Our benchmark identification relies on the first-order conditions (FOCs) of cost minimization for variable inputs. In the main specification, we assume both labor (L) and material (M) are fully flexible inputs. However, as noted in the literature (e.g. De Loecker et al., 2016a), labor might be subject to adjustment costs (e.g., hiring and firing costs) or labor market frictions, rendering it a quasi-fixed input in the short run. If labor is not perfectly flexible, the FOC for labor may not hold with equality, potentially affecting the accuracy of our markup proxy.

We relax the assumption that labor is fully flexible. Instead of using both the labor and material first-order conditions to identify the markup, we identify the markup from the material FOC alone. The estimation results are reported in Table 9, with Columns (1)–(2) using the LP estimation procedure and Columns (3)–(4) using the ACF estimation procedure; within each pair, the IV and benchmark specifications are defined as in our main analysis. Across all four specifications, the export productivity premium remains positive and statistically significant, with $\hat{\beta}_{exp}^{\omega}$ ranging from 0.004 to 0.006 under the preferred IV specification. The production technology continues to display substantial increasing returns to scale, with $S = 1.147$ – 1.166 in total and $S^V = 1.102$ – 1.127 in variable inputs. The demand parameter is also positive and statistically significant throughout: $\hat{\beta}_{exp}^D = 0.058$ (SE = 0.0057) and 0.029 (SE = 0.0029) under LP for the IV and benchmark specifications, and 0.052 (SE = 0.0055) and 0.026 (SE = 0.0027) under ACF. The implied mean firm-level demand effect ranges from 8.3% to 16.5% depending on the specification, in line with our main estimates.

¹⁹See Table B7 for export’s effect on log markup.

Table 9: Robustness to Flexible Variable Inputs

Parameter	(1)	(2)	(3)	(4)
	Ours (LP)		Ours (ACF)	
	(IV)	(Benchmark)	(IV)	(Benchmark)
<i>Panel A: Estimates of Production Function</i>				
β_{exp}^ω (Productivity Effect)	0.006 (0.0021)	0.012 (0.0012)	0.004 (0.0020)	0.010 (0.0012)
β_{exp}^D (Demand Parameter)	0.058 (0.0057)	0.029 (0.0029)	0.052 (0.0055)	0.026 (0.0027)
RTS	1.166 (0.0040)	1.150 (0.0030)	1.166 (0.0039)	1.147 (0.0029)
RTSV	1.123 (0.0042)	1.102 (0.0031)	1.127 (0.0041)	1.104 (0.0031)
α_L	0.045 (0.0008)	0.039 (0.0008)	0.068 (0.0017)	0.062 (0.0015)
α_M	1.077 (0.0042)	1.063 (0.0030)	1.059 (0.0043)	1.042 (0.0027)
α_K	0.044 (0.0009)	0.048 (0.0006)	0.039 (0.0008)	0.043 (0.0006)
<i>Panel B: The Implied Demand Effect of Export</i>				
Mean	0.165	0.087	0.159	0.083
Median	0.151	0.080	0.144	0.075
SD	0.064	0.035	0.065	0.036
IQR	0.085	0.046	0.085	0.046
IDR	0.167	0.092	0.169	0.093
Observations	787,247	1,234,292	787,247	1,234,292

Note 1: This table assesses the robustness of our main results to relaxing the assumption that labor is fully flexible. Instead of using both the labor and material first-order conditions (FOCs) to identify the markup, we identify the markup from the material FOC alone, which remains valid when labor is subject to adjustment frictions or other rigidities. Columns (1)–(2) use the LP and Columns (3)–(4) use the ACF; within each pair, the IV and benchmark specifications follow Tables 3 and 8.

Note 2: For comparability with Tables 3 and 8, S^V is defined as the sum of the labor and material output elasticities ($\alpha_L + \alpha_M$), even though only the material FOC is used to identify the markup.

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across firms with differing market power. The panel reports the mean, median, SD, IQR, and IDR of this firm-level effect after symmetric 10% truncation at both tails.

Note 4: Standard errors clustered at the firm level are reported in parentheses.

Relative to the main results in Tables 3 and 8, identifying the markup from the material FOC alone produces a somewhat larger material output elasticity, a somewhat smaller labor output elasticity, and correspondingly larger estimates of returns to scale. The pattern is intuitive:

when the labor FOC is not imposed, the cost-share information no longer disciplines the labor elasticity, so the joint-estimation procedure attributes more of the revenue variation to materials and to overall scale. Crucially, however, the qualitative conclusions of the paper are preserved. Exports still generate a positive productivity premium and a positive demand effect; production still displays increasing returns to scale; and the contrast between our method and the traditional revenue-based methods is unchanged. We therefore conclude that our main results are robust to relaxing the assumption that labor is fully flexible.

7.4 Translog Production Function Estimation

In the Cobb-Douglas case in our main result, one limitation is that all variation in markup is driven by expenditure shares. This is because output elasticities are constant in the Cobb-Douglas case. We relax this assumption to consider the more flexible translog production function, in which case both expenditure shares and the flexible output elasticities contribute to the changes in markup. In this more flexible case, our key insight of controlling for firm heterogeneity in markup using the raw markup and estimate markup and production function jointly still holds. However, the estimation would be slightly adjusted to estimate all parameters in one step using GMM, as discussed in Section 3.3.

The estimation results are reported in Table B4. Our results are robust, and the translog specification delivers an even larger export premium on productivity. The first-order coefficients are $\alpha_L \approx 0.08$, $\alpha_M \approx 0.81$, and $\alpha_K \approx 0.04$; combined with the second-order coefficients, the sample-mean returns to scale rise to 1.690 (IV) and 1.663 (Benchmark), with variable-input returns 1.584 and 1.552. The productivity premium is $\hat{\beta}_{exp}^\omega = 0.018$ (IV) and 0.007 (Benchmark), and the demand parameter is $\hat{\beta}_{exp}^D = 0.088$ (IV, SE = 0.0101) and 0.107 (Benchmark, SE = 0.0027), with a mean firm-level demand effect of 11.1% and 14.0%, respectively. We also analyze the export effect on markup and find that export raises firms' markup by 1.3-1.7% on average (See Table B5), with a similar dynamic pattern as that in the Cobb-Douglas production function case (See Figure B2). From translog's case, we can see that when the output elasticities are variable, export shows even greater effects on productivity and markup, while the level of IRS and demand shifter effect remain similar. With the flexible translog production function, the increasing returns to scale demonstrate substantial heterogeneity across firms, with a standard deviation of 0.091 and an inter-decile range of 0.220. All other results are qualitatively and quantitatively similar to our main results. This exercise suggests that our methodology can be applied to more general production functions and the main results are robust.

7.5 Validation using External Data

One potential concern is whether the estimated result — in particular, increasing returns to scale and gains from export — is a unique feature that only exists in the Chinese firm-level data. To show that our results are universal and not specific to the Chinese economy, we estimated our model using the plant-level data from Colombia from 1981 to 1991, which reports similar input and output data, although it is at the plant level. For a detailed introduction to the dataset, please refer to [Roberts and Tybout \(1997\)](#), [Grieco et al. \(2016a\)](#), and [Zhang \(2017\)](#). The estimation results are reported in the first two columns in Appendix A.10 Table A3, using different estimation methods. The results are consistent with those from Chinese data. In particular, exporting has a positive and statistically significant effect on productivity. Technology shows strong increasing returns to scale, which contributes to reducing production costs, given that export increases the market size substantially.

As a comparison, columns (3)–(4) report the estimation results using the widely adopted approaches in the literature based on revenue data to estimate the revenue production function without correcting for firm heterogeneity in markup. Specifically, column (3) assumes a constant markup, following [Klette and Griliches \(1996\)](#), while column (4) uses deflated revenue without any markup adjustment. Consistent with our main results from Chinese data, the traditional methods fail to capture the export premium on productivity and market demand. They also substantially underestimate the returns to scale, although in the Colombia data it shows slight increasing returns to scale even using the traditional estimation methods.

We then repeat the gains-from-export decomposition of Section 6 on the Colombian plant-level sample. The results, reported in Table 10, show that exporting reduces marginal cost by 5.894% on average, of which 5.274 percentage points (89.481%) operate through increasing returns to scale and the remaining 0.620 percentage points (10.519%) through the productivity-premium channel. The IRS share is substantially larger than the 61.297% found for Chinese exporters in Table 6, in line with the higher estimated returns to scale in Colombia (1.239 under LP and 1.362 under ACF) than in China. The markup response is also larger (+2% versus +1% in China), the consumer-price decline is correspondingly bigger (−3.894% versus −0.854%), and exporter profits rise by 35.103% (against 25.384% in the Chinese sample). The Colombian exercise thus confirms qualitatively that exporting delivers efficiency gains through both productivity and IRS channels, and shows that the IRS channel is quantitatively even more dominant in the smaller, more export-competing Colombian economy.

In the Chilean data, although [Garcia-Marin and Voigtländer \(2019\)](#) assume constant returns

Table 10: Gains from Export Decomposition (Colombian Data)

Marginal Cost ↓ 5.894%		Markup	Consumer Price	Exporter Profits
<i>through IRS</i>	<i>through Productivity</i>			
↓ 5.274%	↓ 0.620%	↑ 2%	↓ 3.894%	↑ 35.103%

Note 1: This table decomposes the gains from exporting using the Colombian plant-level data over 1981–1991 (see Section 7.5). According to equation (22), exporters’ marginal cost declines by 5.894% on average within the first year of exporting. Of this decline, 89.481% (5.274 percentage points) operates through increasing returns to scale (IRS), and the remaining 10.519% (0.620 percentage points) is attributable to the export productivity premium.

Note 2: The markup rises by 2% on average upon exporting. Combining the markup increase with the marginal-cost decline through the decomposition in equation (23) implies that the consumer price falls by 3.894% on average.

Note 3: Applying the decomposition in equation (26), exporter profits increase by 35.103% on average.

Note 4: Compared with the Chinese decomposition in Table 6 referring to Table B9, exports generate substantially larger cost savings in Colombia: marginal cost falls by 5.894% versus 1.854% in China, and consumer prices fall by 3.894% versus 0.854%. The IRS channel dominates in both economies but accounts for a much larger share in Colombia (89.481% versus 61.297%), reflecting the stronger increasing returns documented in Table A3. The productivity-premium channel is similar in absolute size (0.620 versus 0.720 percentage points), so the wider total cost savings in Colombia are almost entirely driven by the IRS channel. The markup response is also larger (+2% versus +1%), translating into a bigger profit gain for new exporters (+35.103% versus +25.384%).

to scale in their analysis, their results indicate that the mean returns to scale of Chilean manufacturing industries is 1.023, estimated using output quantity data. Approximately 70% of the industries in their sample exhibit increasing returns to scale. While they discuss the potential bias in the estimated export productivity premium when technology displays increasing returns to scale, they do not explicitly treat increasing returns to scale as a significant source of efficiency gains from export. Moreover, their estimates of returns to scale are likely downward-biased, as they do not account for heterogeneity in input prices. As demonstrated in Section 5, estimating the production function based solely on output quantities without considering input price heterogeneity tends to underestimate returns to scale.

This result of increasing returns to scale also aligns with the estimates of De Loecker et al. (2016a) using physical output quantity data from India. They find that approximately 68% of their sample from India exhibits increasing returns to scale after adjusting for potential biases introduced by input prices and quality. Similarly, Lu and Yu (2015) utilize the same output quantity data as ours and find an average returns to scale of 1.125 among Chinese manufacturing firms after accounting for input price biases.

8 Conclusion

The productivity effect of exports has been the foundation for many trade policies. However, empirical studies typically detected limited/mixed productivity effects of exports using revenue data, even when exports increase markup. We show that this is because export

may reduce output prices, due to efficiency gains from increased productivity and, more importantly, increasing returns to scale. Because output prices are typically unavailable in most micro datasets, we develop a new method to consistently estimate firm-level markup, productivity, and returns to scale jointly, using revenue and widely available variable inputs expenditure. The new method only requires cost minimization, but not profit maximization, as in [De Loecker and Warzynski \(2012\)](#).

We find three main results in the application to Chinese manufacturing industries. First, exports generate large efficiency gains, reducing marginal production costs by about 1.85% on average. Nearly two thirds (61.3%) of the efficiency gain is contributed by increasing returns to scale; the remainder is due to improved productivity after export. Second, improved production efficiency and a larger market allow exporters to charge a higher markup (by about 1%) at a lower price. Altogether, firm profit increases by about 25.4% after export in the Chinese manufacturing industry; consumers also benefit from lower prices, by about 0.85%. The estimation results are validated using a sub-sample with the output quantity information by estimating the physical output production function which is not affected by the unobserved markup heterogeneity. The results are also robust to alternative estimation methods, more flexible specification of the production functions, industry-separated estimation, and export instruments.

Our results suggest that the export productivity premium does exist when correctly estimated by dealing with the unobserved prices. After correcting for unobserved markup, production shows significant increasing returns to scale, which serves as an important source of efficiency gains from export, even at the firm level. We show that unlike [Garcia-Marin and Voigtländer \(2019\)](#), the TFPR effect of export may be limited even when markup increases in the presence of increasing returns to scale in production. The results have implications in many other fields related to firm growth, industrial dynamics, macro growth, and international trade.

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A Appendix - Derivations of the Model

A.1 Markup Derivation for Cobb-Douglas Production Function

Firm j 's cost minimization problem can be described as:

$$\begin{aligned} \min_{\{L_{jt}, M_{jt}\}} \quad & P_{jt}^L L_{jt} + P_{jt}^M M_{jt} \\ \text{s.t.} \quad & e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K} \geq Y_{jt}. \end{aligned} \quad (\text{A1})$$

The corresponding Lagrangian function is:

$$\mathcal{L} = P_{jt}^L L_{jt} + P_{jt}^M M_{jt} + \lambda_{jt} [Y_{jt} - e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K}]$$

The First Order Condition w.r.t. L_{jt} and M_{jt} are shown as follows:

$$P_{jt}^L = \lambda_{jt} \frac{\partial Y_{jt}(\cdot)}{\partial L_{jt}}, \quad (\text{A2})$$

$$P_{jt}^M = \lambda_{jt} \frac{\partial Y_{jt}(\cdot)}{\partial M_{jt}}, \quad (\text{A3})$$

where $Y_{jt}(\cdot)$ represents the production function. Multiplying both sides by L_{jt} and M_{jt} on the above two equations, respectively, and adding them together yield:

$$P_{jt}^L L_{jt} + P_{jt}^M M_{jt} = \lambda_{jt} \left[\frac{\partial Y_{jt}(\cdot)}{\partial L_{jt}} L_{jt} + \frac{\partial Y_{jt}(\cdot)}{\partial M_{jt}} M_{jt} \right].$$

Dividing revenue R_{jt} ($R_{jt} = P_{jt} Y_{jt}$) by both sides yields:

$$\frac{R_{jt}}{P_{jt}^L L_{jt} + P_{jt}^M M_{jt}} = \frac{P_{jt}}{(\alpha_L + \alpha_M) \lambda_{jt}}.$$

λ_{jt} is the shadow price of producing one unit of output, representing the marginal cost $\lambda_{jt} \equiv MC_{jt}$. By definition, the markup $\mu_{jt} = P_{jt}/MC_{jt}$. We can get the expression for firm markup μ_{jt} as follows:

$$\begin{aligned} \mu_{jt} &= (\alpha_L + \alpha_M) \cdot \frac{R_{jt}}{P_{jt}^L L_{jt} + P_{jt}^M M_{jt}} \\ &= (\alpha_L + \alpha_M) \cdot \tilde{\mu}_{jt} \end{aligned} \quad (\text{A4})$$

A.2 Profit Maximization under Effective Demand

This appendix provides the formal proof that the demand-appeal framework, which specifies demand over “effective quantities” and “effective prices” following [Feenstra and Romalis \(2014\)](#), strictly preserves the standard Lerner-index formula for the physical markup. This result validates the structural substitution $\mu_{jt} = \eta_{jt}/(1 + \eta_{jt})$ used in our main empirical

equations.

Recall the effective demand function specified in equation (6):

$$\frac{\tilde{Y}_{jt}}{\bar{Y}_{Jt}} = \left(\frac{\tilde{P}_{jt}}{\bar{P}_{Jt}} \right)^{\eta_{jt}},$$

where the effective quantity and effective price are defined as $\tilde{Y}_{jt} = Y_{jt}e^{z_{jt}}$ and $\tilde{P}_{jt} = P_{jt}/e^{z_{jt}}$, respectively.

Step 1: Mapping Effective Demand to Physical Demand. Substituting the physical variables into the effective demand function yields

$$Y_{jt}e^{z_{jt}} = \bar{Y}_{Jt} \left(\frac{P_{jt}/e^{z_{jt}}}{\bar{P}_{Jt}} \right)^{\eta_{jt}} = \bar{Y}_{Jt} \bar{P}_{Jt}^{-\eta_{jt}} P_{jt}^{\eta_{jt}} e^{-z_{jt}\eta_{jt}}.$$

Solving for the physical quantity Y_{jt} ,

$$Y_{jt} = \left[\bar{Y}_{Jt} \bar{P}_{Jt}^{-\eta_{jt}} e^{-z_{jt}(1+\eta_{jt})} \right] P_{jt}^{\eta_{jt}}.$$

Grouping all the bracketed terms—which are exogenous to the firm's own pricing decision P_{jt} —into a composite demand shifter Φ_{jt} , we obtain

$$Y_{jt} = \Phi_{jt} P_{jt}^{\eta_{jt}}.$$

Step 2: Profit Maximization in Physical Units. The firm maximizes its profit by choosing the physical price P_{jt} , facing the physical cost function $C(Y_{jt})$:

$$\max_{P_{jt}} \Pi_{jt} = P_{jt}Y_{jt} - C(Y_{jt}).$$

Substituting $Y_{jt} = \Phi_{jt}P_{jt}^{\eta_{jt}}$ gives

$$\max_{P_{jt}} \Pi_{jt} = \Phi_{jt}P_{jt}^{1+\eta_{jt}} - C(\Phi_{jt}P_{jt}^{\eta_{jt}}),$$

and the first-order condition with respect to P_{jt} is

$$(1 + \eta_{jt})\Phi_{jt}P_{jt}^{\eta_{jt}} - \underbrace{C'(Y_{jt})}_{MC_{jt}} \cdot \eta_{jt}\Phi_{jt}P_{jt}^{\eta_{jt}-1} = 0.$$

Dividing by $\Phi_{jt}P_{jt}^{\eta_{jt}-1}$ (which equals Y_{jt}/P_{jt} and is strictly positive) yields

$$(1 + \eta_{jt})P_{jt} - MC_{jt} \cdot \eta_{jt} = 0,$$

and rearranging gives the standard physical-markup expression

$$\mu_{jt} \equiv \frac{P_{jt}}{MC_{jt}} = \frac{\eta_{jt}}{1 + \eta_{jt}}.$$

Economic Intuition. The derivation shows that introducing an exponential demand shifter $e^{z_{jt}}$ does not alter the classic markup formula. In our log-linear/VES demand framework, the demand-appeal shock z_{jt} acts purely as a vertical demand shifter that is fully absorbed into the intercept Φ_{jt} . While it changes the absolute physical quantity demanded at any given price, it does not alter the consumer’s marginal sensitivity to the physical price—the physical price elasticity remains exactly η_{jt} . As long as the physical price elasticity is η_{jt} , the optimal markup over marginal cost is uniquely pinned down by this elasticity, formally justifying the integration of the structural demand system with the production approach in the spirit of [De Loecker and Warzynski \(2012\)](#).

A.3 Timing Assumption and Identification

The estimation procedure in Section 3.2 resolves the identification challenges in the production approach to markup estimation when only revenues are observed, but not output prices. Below, we (i) lay out the information-timing assumption that makes our control-function strategy valid, (ii) verify that the resulting equation (11) survives the recent critiques of [Doraszelski and Jaumandreu \(2024\)](#) and [Bond et al. \(2021\)](#), and (iii) address the endogeneity of export status due to firms’ self-selection into foreign markets.

Information Timing Assumption. A central feature of our empirical framework is the timing of information, which follows an “*ex-ante production, ex-post pricing*” sequence:

1. **Ex-ante Production.** In period t , firm j observes its structural productivity ω_{jt} and export status D_{jt}^{exp} , but does not observe the idiosyncratic demand-appeal shock ϵ_{jt}^D . Based on these observations and expectations, the firm chooses flexible inputs and produces a fixed physical output Y_{jt} .
2. **Shock Realization.** After physical production is completed, the idiosyncratic demand shock ϵ_{jt}^D is revealed to the firm and consumers.
3. **Ex-post Pricing (Market Clearing).** With the physical output Y_{jt} fixed in the short run, the firm observes ϵ_{jt}^D and sets the optimal market-clearing price P_{jt} .²⁰

This sequential timing resolves the identification critique of [Doraszelski and Jaumandreu \(2024\)](#), where substituting variable markups into revenue equations typically creates intractable endogenous interactions between μ_{jt} and unobserved shocks. The validity of our control-function approach hinges on three econometric properties that follow directly from this timing

²⁰The ex-post timing of pricing is essential for the markup identity (10) to hold pointwise in observed data. With P_{jt} clearing the market at the ex-ante produced Y_{jt} , realized revenue equals $P_{jt}Y_{jt}$, so the cost-minimization FOCs combined with the raw markup deliver $\mu_{jt} = S^V \tilde{\mu}_{jt}$. Under ex-ante pricing, realized quantity sold would generally differ from the produced Y_{jt} (creating excess demand or inventory), so the identity would hold only in expectation and $\tilde{\mu}_{jt}$ would be a noisy proxy for μ_{jt} .

assumption.

Scalar Unobservability. For simplicity, we ignore the additional labor adjustment error required by the identification in LP estimation procedure²¹ and the industry-level output indice $\bar{y}_{Jt}$. Following Akerberg (2016), we assume labor is subject to adjustment costs and is chosen prior to materials. When the firm chooses its intermediate input m_{jt} , the idiosyncratic demand shock ϵ_{jt}^D has not yet been realized. Therefore, the firm's optimal material demand depends only on the ex-ante state variables $(l_{jt}, k_{jt}, D_{jt}^{exp})$ and a single unobserved scalar—structural productivity ω_{jt} :

$$m_{jt} = m(l_{jt}, k_{jt}, \omega_{jt}, D_{jt}^{exp}).$$

Material demand is strictly independent of the ex-post shock ϵ_{jt}^D . Had the firm observed ϵ_{jt}^D before its input choices, material demand would depend on two unobservables, rendering the inversion impossible.

Strict Invertibility. Given an expected demand curve, a higher structural productivity ω_{jt} reduces the firm's marginal cost, leading to a higher planned optimal physical output, which strictly requires more material inputs. Because $\partial m_{jt} / \partial \omega_{jt} > 0$ holds strictly, we can validly invert material demand to non-parametrically control for productivity, $\omega_{jt} = m^{-1}(m_{jt}, l_{jt}, k_{jt}, D_{jt}^{exp})$, as used in (12).

Strict Exogeneity and Elimination of Interactions. Ex-post, the unobserved demand shock ϵ_{jt}^D shifts the market-clearing price. In a standard setup, substituting variable markups into the revenue equation creates intractable endogenous interactions (e.g., $(1/\mu_{jt})\epsilon_{jt}^D$). However, our framework circumvents this critique. By multiplying (8) by the observable proxy $\tilde{\mu}_{jt}$ (derived from the firm's first-order conditions), the non-linear unobservable interactions perfectly cancel out: all endogenous price variations are folded into the dependent variable $\tilde{\mu}_{jt}(r_{jt} - \bar{r}_{Jt})$, while the right-hand side of (11) features only ex-ante chosen inputs and a strictly additive error term with the constant coefficient $1/(\alpha_L + \alpha_M)$. Thus, the standard orthogonality condition

$$\mathbb{E}[\epsilon_{jt}^D \mid l_{jt}, m_{jt}, k_{jt}, D_{jt}^{exp}] = 0$$

holds by our timing assumption, allowing for consistent structural estimation.

A.4 The Second Stage Estimation

Recall the AR(1) productivity evolution process:

$$\omega_{jt} = \beta_0^\omega + \beta_1^\omega \omega_{jt-1} + \beta_{exp}^\omega D_{jt}^{exp} + \epsilon_{jt}^\omega. \quad (\text{A5})$$

where ϵ_{jt}^ω is uncorrelated with ω_{jt-1} and D_{jt}^{exp} . Thus by the formula of the OLS estimator, the parameters can be expressed as:

$$[\beta_0^\omega \quad \beta_1^\omega \quad \beta_{exp}^\omega]' = (X'_{jt-1} X_{jt-1})^{-1} X'_{jt-1} \omega_{jt}, \quad (\text{A6})$$

²¹For details, please refer to Akerberg et al. (2015).

where X_{jt-1} is the vector given by $X_{jt-1} = [1 \ \omega_{jt-1} \ D_{jt-1}^{\text{exp}}]$. Using equation (13) to substitute ω_{jt} and ω_{jt-1} in equation (A6), and combine the resulting equations (A6) and (14). Then, in the second step, we could express the ϵ_{jt}^ω term as a function of parameters to be estimated $\epsilon_{jt}(\alpha_M, \alpha_K, \beta_{exp}^D)$ and adopt the simple General Moment Method with moment conditions:

$$E \left(\epsilon_{jt}(\alpha_M, \alpha_K, \beta_{exp}^D) \begin{pmatrix} m_{jt-1} \\ k_{jt} \\ \tilde{\mu}_{jt-1} D_{jt-1}^{exp} \end{pmatrix} \right) = 0. \quad (\text{A7})$$

With the estimates of $\hat{\alpha}$ estimated in the first step, we can get the full estimation of all the parameters.

A.5 Markup Derivation for Translog Production Function

Firm j 's optimization problem can be described as:

$$\begin{aligned} \min_{\{L_{jt}, M_{jt}\}} & P_{jt}^L L_{jt} + P_{jt}^M M_{jt} \\ \text{s.t.} & e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K} L_{jt}^{\alpha_{LL} l_{jt}} M_{jt}^{\alpha_{MM} m_{jt}} K_{jt}^{\alpha_{KK} k_{jt}} L_{jt}^{\alpha_{LM} m_{jt}} M_{jt}^{\alpha_{MK} k_{jt}} L_{jt}^{\alpha_{LK} k_{jt}} \geq Y_{jt}. \end{aligned} \quad (\text{A8})$$

The corresponding Lagrangian function is:

$$\begin{aligned} \mathcal{L} = & P_{jt}^L L_{jt} + P_{jt}^M M_{jt} + \lambda_{jt} [Y_{jt} \\ & - e^{\omega_{jt}} L_{jt}^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K} L_{jt}^{\alpha_{LL} l_{jt}} M_{jt}^{\alpha_{MM} m_{jt}} K_{jt}^{\alpha_{KK} k_{jt}} L_{jt}^{\alpha_{LM} m_{jt}} M_{jt}^{\alpha_{MK} k_{jt}} L_{jt}^{\alpha_{LK} k_{jt}}] \end{aligned}$$

The First Order Conditions w.r.t. L_{jt} and M_{jt} are shown as follows:

$$\begin{aligned} P_{jt}^L &= \lambda_{jt} \frac{\partial Y_{jt}(\cdot)}{\partial L_{jt}}, \\ P_{jt}^M &= \lambda_{jt} \frac{\partial Y_{jt}(\cdot)}{\partial M_{jt}}, \end{aligned}$$

where $Y_{jt}(\cdot)$ represents the translog production function. Multiplying both sides by L_{jt} and M_{jt} on the above two equations, respectively, and adding them together yield:

$$P_{jt}^L L_{jt} + P_{jt}^M M_{jt} = \lambda_{jt} \left[\frac{\partial Y_{jt}(\cdot)}{\partial L_{jt}} L_{jt} + \frac{\partial Y_{jt}(\cdot)}{\partial M_{jt}} M_{jt} \right].$$

Dividing revenue R_{jt} ($R_{jt} = P_{jt} Y_{jt}$) by both sides yields:

$$\frac{R_{jt}}{P_{jt}^L L_{jt} + P_{jt}^M M_{jt}} = \frac{P_{jt}}{(\alpha_L^* + \alpha_M^*) \lambda_{jt}},$$

where $\alpha_L^* \equiv \alpha_L + 2\alpha_{LL} l_{jt} + \alpha_{LM} m_{jt} + \alpha_{LK} k_{jt}$, $\alpha_M^* \equiv \alpha_M + 2\alpha_{MM} m_{jt} + \alpha_{LM} l_{jt} + \alpha_{MK} k_{jt}$. λ_{jt} is the shadow price of producing one unit of output, representing the marginal cost $\lambda_{jt} \equiv MC_{jt}$. By definition, the markup $\mu_{jt} = P_{jt}/MC_{jt}$. We can get the expression for firm markup μ_{jt} as

follows:

$$\begin{aligned}\mu_{jt} &= (\alpha_L^* + \alpha_M^*) \cdot \frac{R_{jt}}{P_{jt}^L L_{jt} + P_{jt}^M M_{jt}} \\ &= (\alpha_L^* + \alpha_M^*) \cdot \tilde{\mu}_{jt}\end{aligned}\tag{A9}$$

A.6 Discussions on the Potential Underlining Market Structure for the VES Demand Function

This appendix provides one explicit micro-foundation for the VES inverse demand function (6) used in Section 3. Starting from Cobb–Douglas household preferences across industries, CES demand across product varieties within an industry, and Bertrand competition across firms producing a given variety, we derive a firm-level inverse demand of the form²²

$$\frac{P_{jt}}{\exp(z_{jt})} = \left[\frac{Y_{jt} \exp(z_{jt})}{\bar{Y}_{Jt}} \right]^{1/\eta_{jt}} \bar{P}_{Jt},\tag{A10}$$

in which the firm-time-specific demand elasticity $\eta_{jt} = -[S_{jt}^k \sigma + (1 - S_{jt}^k) \rho]$ depends on the firm’s market share S_{jt}^k in its product variety k , and the demand appeal $z_{jt} = \beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D$ enters as a vertical shifter on effective quantities and effective prices, exactly as in the main text. Throughout, P_{jt} denotes the physical price; the mapping to the effective price used in the main text is $\tilde{P}_{jt} = P_{jt} / \exp(z_{jt})$. The Lerner-index relation $\mu_{jt} = \eta_{jt} / (1 + \eta_{jt})$ that we use throughout the empirical analysis is preserved under this effective-demand reformulation; see Appendix A.2. Related micro-foundations have been studied by [Atkeson and Burstein \(2008\)](#), [Edmond et al. \(2015\)](#), [Arkolakis and Morlacco \(2017\)](#), and [Edmond et al. \(2023\)](#).

The economy has three nested layers — industries J , product varieties k within each industry, and a finite number of firms j within each variety. Each firm produces under one variety and one industry. The market structure is monopolistic competition across varieties and Bertrand oligopoly within a variety, so that the firm’s market share S_{jt}^k is decreasing in its own price and its markup is increasing in its market share. Marginal-cost declines after exporting (driven by the productivity premium and IRS) therefore translate into lower prices, larger market shares, and higher markups — the mechanism documented empirically in the main text.

A.6.1 Households

As in [Caliendo and Parro \(2015\)](#), we assume there are $J = 1, \dots, N$ industries and representative households maximize the utility by consuming final goods Y_{Jt} . The preferences of the

²²The mapping from the exact physical-price form derived below to the effective-price form (A10) requires the first-order log-linearization around the symmetric-firms benchmark discussed at the end of this appendix.

households are given by:

$$U_t = \prod_{J=1}^N C_{Jt}^{\alpha_J}, \text{ where } \sum_{J=1}^N \alpha_J = 1, \quad (\text{A11})$$

The market between final goods C_{Jt} is assumed to be perfectly competitive, so the final goods producers don't have the market power and cannot decide the price. Since the utility is in the Cobb-Douglas form, the market share of final goods J is fixed and equals to α_J .

A.6.2 Final Goods: Monopolistic Competition

Similar to [Akcigit and Ates \(2023\)](#)²³, producers of final goods in industry J supply Y_{Jt} , $Y_{Jt} = C_{Jt}$ by market clearing condition, according to the following production technology:

$$Y_{Jt} = \left[\int_0^1 Y_{kt}^{1-\frac{1}{\sigma}} dk \right]^{\frac{\sigma}{\sigma-1}}, \quad (\text{A12})$$

where Y_{kt} denotes the amount of intermediate variety $k \in [0, 1]$, and σ is the constant elasticity of substitution between intermediate varieties. The market for intermediate varieties is monopolistic competition. So as is standard, the theoretical price index $\bar{P}_{Jt}$ for the final goods is given by:

$$\bar{P}_{Jt} = \left[\int_0^1 (P_{kt})^{1-\sigma} dk \right]^{\frac{1}{1-\sigma}}. \quad (\text{A13})$$

The product-industry relative demand functions for the output are given by:

$$Y_{kt} = \left(\frac{P_{kt}}{\bar{P}_{Jt}} \right)^{-\sigma} Y_{Jt} \quad (\text{A14})$$

A.6.3 Intermediate Goods: Oligopoly

Within the intermediate product variety k in industry J is oligopoly (with Bertrand Competition). In each industry J and product variety k , there are N_{kt} firms selling the same kind of goods. Output in each intermediate product variety k within industry J is given by a CES production function:

$$Y_{kt} = \left[\sum_{j=1}^{N_{kt}} (Y_{jt})^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{A15})$$

Where Y_{jt} denotes the output of firm j in product variety k in the industry J , and ρ is the constant elasticity of substitution between firms' production. As in [Atkeson and Burstein \(2008\)](#), we assume that firms' goods are imperfect substitutes, and goods within a variety are more substitutable than goods across varieties, i.e., $1 < \sigma < \rho < \infty$. As is standard, the

²³The difference is that we change the technology here to be CES.

theoretical price index P_{kt} for the product variety k is given by:

$$P_{kt} = \left[\sum_{j=1}^{N_{kt}} (P_{jt})^{1-\rho} \right]^{\frac{1}{1-\rho}}. \quad (\text{A16})$$

The firm-product relative demand functions for the output are given by:

$$Y_{jt} = \left(\frac{P_{jt}}{P_{kt}} \right)^{-\rho} Y_{kt}. \quad (\text{A17})$$

We also know that the market share of firm j in product k is given by:

$$S_{jt}^k = \frac{P_{jt} Y_{jt}}{P_{kt} Y_{kt}} = \left(\frac{Y_{jt}}{Y_{kt}} \right)^{-\frac{1}{\rho}} \frac{Y_{jt}}{Y_{kt}} \quad (\text{A18})$$

$$= \left(\frac{Y_{jt}}{Y_{kt}} \right)^{\frac{\rho-1}{\rho}} \quad (\text{A19})$$

$$= \left(\frac{P_{jt}}{P_{kt}} \right)^{1-\rho}. \quad (\text{A20})$$

Differentiating the closed-form share $S_{jt}^k = P_{jt}^{1-\rho} / \sum_{i=1}^{N_{kt}} P_{it}^{1-\rho}$ with respect to P_{jt} gives $\partial S_{jt}^k / \partial P_{jt} = -(\rho - 1) S_{jt}^k (1 - S_{jt}^k) / P_{jt} \leq 0$: the firm's market share is strictly decreasing in its own price.

A.6.4 Market Structure: VES Demand Function

As in [Atkeson and Burstein \(2008\)](#), we assume that firms play a static game but with a price (Bertrand) competition.²⁴ Specifically, each firm j chooses its price P_{jt} for product variety k which belongs to industry J taking as given the prices chosen by the other firms in the economy, as well as the input prices and the industry price $\bar{P}_{Jt}$ and quantity Y_{Jt} . Note that under this assumption, each firm does recognize that product prices P_{kt} and quantities Y_{kt} vary when that firm changes its quantity Y_{jt} .

Then, firm j in producing product k which belongs industry J solve the profit maximization at time t :

$$\max_{P_{jt}, Y_{jt}} P_{jt} Y_{jt} - \text{Cost}(Y_{jt}) \quad (\text{A21})$$

$$\text{s.t. } Y_{jt} = \left(\frac{P_{jt}}{P_{kt}} \right)^{-\rho} \left(\frac{P_{kt}}{\bar{P}_{Jt}} \right)^{-\sigma} Y_{Jt} \quad (\text{A22})$$

Equation (A22) is derived from combining equation (A14) and equation (A17), which implies how the firm-industry relative demand is influenced by the firm-product relative price and

²⁴As discussed in [Arkolakis and Morlacco \(2017\)](#), the Cournot competition case is qualitatively the same as the Bertrand case. Another derivation method of Bertrand competition can also be found in [Amiti et al. \(2019\)](#).

the product-industry relative price. However, neither the ρ nor σ represents the demand elasticity. To derive the demand elasticity, we first rewrite the profit function in the form of:

$$\pi_{jt} = P_{jt} \left(\frac{P_{jt}}{P_{kt}} \right)^{-\rho} \left(\frac{P_{kt}}{\bar{P}_{Jt}} \right)^{-\sigma} Y_{Jt} - \text{Cost}(Y_{jt}) \quad (\text{A23})$$

$$= P_{jt}^{1-\rho} P_{kt}^{\rho-\sigma} \bar{P}_{Jt}^{\sigma} Y_{Jt} - \text{Cost}(Y_{jt}) \quad (\text{A24})$$

$$= P_{jt}^{1-\rho} \left\{ \left[\sum_{j=1}^{N_{kt}} (P_{jt})^{1-\rho} \right]^{\frac{1}{1-\rho}} \right\}^{\rho-\sigma} \bar{P}_{Jt}^{\sigma} Y_{Jt} - \text{Cost}(Y_{jt}). \quad (\text{A25})$$

And the FOC of P_{jt} gives:

$$\partial \pi_{jt} / \partial P_{jt} = 0 \quad (\text{A26})$$

$$\begin{aligned} \Rightarrow (1-\rho) P_{jt}^{-\rho} \left[\sum_{j=1}^{N_{kt}} (P_{jt})^{1-\rho} \right]^{\frac{\rho-\sigma}{1-\rho}} \bar{P}_{Jt}^{\sigma} Y_{Jt} + \frac{\rho-\sigma}{1-\rho} \left[\sum_{j=1}^{N_{kt}} (P_{jt})^{1-\rho} \right]^{\frac{\rho-\sigma}{1-\rho}-1} (1-\rho) P_{jt}^{-\rho} P_{jt}^{1-\rho} \bar{P}_{Jt}^{\sigma} Y_{Jt} \\ = MC_{jt} \frac{\partial \left(\frac{P_{jt}}{P_{kt}} \right)^{-\rho} \left(\frac{P_{kt}}{\bar{P}_{Jt}} \right)^{-\sigma} Y_{Jt}}{\partial P_{jt}} \end{aligned} \quad (\text{A27})$$

$$\Rightarrow (1-\rho) Y_{jt} + (\rho-\sigma) \left(\frac{P_{jt}}{P_{kt}} \right)^{1-\rho} Y_{jt} = MC_{jt} \left[-\rho \frac{Y_{jt}}{P_{jt}} + (\rho-\sigma) \left(\frac{P_{jt}}{P_{kt}} \right)^{1-\rho} \frac{Y_{jt}}{P_{jt}} \right] \quad (\text{A28})$$

$$\Rightarrow \frac{P_{jt}}{MC_{jt}} = \frac{\rho - (\rho - \sigma) \left(\frac{P_{jt}}{P_{kt}} \right)^{1-\rho}}{(\rho - 1) - (\rho - \sigma) \left(\frac{P_{jt}}{P_{kt}} \right)^{1-\rho}} \quad (\text{A29})$$

Since we know that $(P_{jt}/P_{kt})^{1-\rho} = S_{jt}^k$, and markup is defined as $\eta_{jt}/(1 + \eta_{jt})$.²⁵ So we can derive the variable demand elasticity:

$$\mu_{jt} = \frac{\eta_{jt}}{1 + \eta_{jt}} = \frac{\rho - (\rho - \sigma) S_{jt}^k}{(\rho - 1) - (\rho - \sigma) S_{jt}^k} \quad (\text{A30})$$

$$\Rightarrow \eta_{jt} = - [S_{jt}^k \sigma + (1 - S_{jt}^k) \rho] \quad (\text{A31})$$

As discussed in [Arkolakis and Morlacco \(2017\)](#), the markup is increasing in S_{jt}^k . Combined with the result that S_{jt}^k is decreasing in price, marginal-cost declines after exporting (through the productivity premium and IRS) generate higher market shares and, in turn, higher markups under Bertrand competition. Mapping to the notation of Section 3, η_{jt} derived here is precisely the firm-time-specific demand elasticity used in (6), and the implied markup $\mu_{jt} = \eta_{jt}/(1 + \eta_{jt})$ coincides with the structural markup used in (10). The Lerner relation is preserved once we map physical prices to effective prices via $\tilde{P}_{jt} = P_{jt}/\exp(z_{jt})$; see

²⁵This definition of markup does not rely on monopolistic competition. Because the general demand elasticity is defined by $\eta \equiv (dY/Y)/(dP/P) = (dY/dP) \cdot (P/Y)$. From the F.O.C. of the profit maximization $\max_P \{PY - \text{Cost}(Y)\}$, s.t. $Y = D(P)$, we will always have $Y + P \cdot (dY/dP) = MC \cdot (dY/dP) \Rightarrow Y + \eta \cdot Y = MC \cdot \eta \cdot (Y/P) \Rightarrow P/MC = \eta/(1 + \eta)$.

Appendix A.2.

From the CES–Bertrand Demand to the Empirical Demand Function

Substituting $\rho = -[(\eta_{jt} + S_{jt}^k \sigma)/(1 - S_{jt}^k)]$ into (A22) and simplifying yields the exact CES–Bertrand inverse demand

$$y_{jt} = \eta_{jt} p_{jt} - \eta_{jt} \bar{p}_{Jt} + \bar{y}_{Jt} + \bar{y}_{jt}, \quad (\text{A32})$$

where

$$\bar{y}_{jt} \equiv \frac{\rho - \sigma}{\rho \sigma} [S_{jt}^k \sigma \ln(Y_{jt}/Y_{kt}) + (1 - S_{jt}^k) \rho \ln(Y_{Jt}/Y_{kt})]$$

is a weighted average of within- and across-variety log relative quantities.²⁶ Our empirical specification in Section 3 replaces $\bar{y}_{jt}$ with the parsimonious linear form $-(1 + \eta_{jt})z_{jt} = -(1 + \eta_{jt})[\beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D]$, collapsing the unobserved relative-quantity object into a deterministic export-driven mean component plus an idiosyncratic ex-post shock. The resulting log-demand specification used in estimation is

$$y_{jt} = \eta_{jt} p_{jt} - \eta_{jt} \bar{p}_{Jt} + \bar{y}_{Jt} - (1 + \eta_{jt})[\beta_{exp}^D D_{jt}^{exp} + \epsilon_{jt}^D]. \quad (\text{A33})$$

This is a first-order log-linearization of (A32) around a symmetric-firms benchmark, in the spirit of Edmond et al. (2015) and the Kimball-aggregator literature: at the symmetric point $S_{jt}^k = 1/N_{kt}$ and $\bar{y}_{jt}$ collapses to a constant absorbed into $\bar{y}_{Jt}$. Two econometric properties make this approximation harmless within our framework. First, by the ex-ante / ex-post timing developed in Section A.3, ϵ_{jt}^D realizes only after physical production is chosen, so any deviation of $\bar{y}_{jt}$ from its symmetric-point value that is absorbed into ϵ_{jt}^D is mechanically orthogonal to the input-choice decision. Second, the markup-adjusted revenue equation (11) loads on ϵ_{jt}^D only through the strictly additive, constant-coefficient term $\epsilon_{jt}^D/(\alpha_L + \alpha_M)$, so heterogeneity in $\bar{y}_{jt}$ does not interact with the estimated production parameters. The economic content of the approximation is the same as in monopolistic-competition implementations of the Kimball aggregator: in equilibrium, the relevant variation in the demand shifter is summarized by z_{jt} , with the residual treated as a strictly additive ex-post shock.

A.7 Marginal Cost Derivation and Decomposition of TFPR

Combining (A2) and (A3) yields :

$$\frac{L_{jt}}{M_{jt}} = \frac{\alpha_L P_{jt}^M}{\alpha_M P_{jt}^L}. \quad (\text{A34})$$

²⁶We aggregate to the industry level J because the revenue dataset does not identify product variety k at the firm level.

Substituting it into the Cobb-Douglas production function yields the expression of M_{jt} :

$$\begin{aligned} Y_{jt} &= e^{\omega_{jt}} \left(\frac{\alpha_L P_{jt}^M}{\alpha_M P_{jt}^L} M_{jt} \right)^{\alpha_L} M_{jt}^{\alpha_M} K_{jt}^{\alpha_K} \\ \Rightarrow M_{jt} &= Y_{jt}^{\frac{1}{\alpha_L + \alpha_M}} \left[e^{\omega_{jt}} K_{jt}^{\alpha_K} \left(\frac{\alpha_L P_{jt}^M}{\alpha_M P_{jt}^L} \right)^{\alpha_L} \right]^{\frac{-1}{\alpha_L + \alpha_M}}. \end{aligned} \quad (\text{A35})$$

The total variable costs function can be expressed as:

$$\begin{aligned} \text{Total Variable Costs}_{jt} &= P_{jt}^L L_{jt} + P_{jt}^M M_{jt} \\ &= P_{jt}^L \left(\frac{\alpha_L P_{jt}^M}{\alpha_M P_{jt}^L} M_{jt} \right) + P_{jt}^M M_{jt} \\ &= \frac{\alpha_L + \alpha_M}{\alpha_M} P_{jt}^M M_{jt}. \end{aligned} \quad (\text{A36})$$

Substituting equation (A35) in (A36), and taking the derivative of Y_{jt} , we get the expression of marginal costs in the natural logarithmic form as follows:

$$\begin{aligned} mc_{jt} &= \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} y_{jt} - \frac{1}{\alpha_L + \alpha_M} \omega_{jt} + \frac{\alpha_M}{\alpha_L + \alpha_M} p_{jt}^M + \frac{\alpha_L}{\alpha_L + \alpha_M} p_{jt}^L \\ &\quad - \frac{\alpha_M}{\alpha_L + \alpha_M} \ln(\alpha_M) - \frac{\alpha_L}{\alpha_L + \alpha_M} \ln(\alpha_L) - \frac{\alpha_K}{\alpha_L + \alpha_M} k_{jt}, \end{aligned} \quad (\text{A37})$$

where the lowercase variables represent the natural logarithmic form of the original variables. We fix and abstract away changes in input prices and capital stock. So the changes in marginal costs can be expressed in the following form:

$$\Delta mc = \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} \Delta y - \frac{1}{\alpha_L + \alpha_M} \Delta \omega. \quad (\text{A38})$$

Then, the changes of TFPR can be decomposed as:

$$\begin{aligned} \Delta \text{TFPR} &= \Delta p + \Delta \omega = \Delta \ln \mu + \Delta mc + \Delta \omega \\ &= \Delta \ln \mu + \frac{1 - (\alpha_L + \alpha_M)}{\alpha_L + \alpha_M} \Delta y - \frac{1}{\alpha_L + \alpha_M} \Delta \omega + \Delta \omega \\ &= \Delta \ln \mu + \left(1 - \frac{1}{\alpha_L + \alpha_M} \right) (\Delta \omega - \Delta y). \end{aligned} \quad (\text{A39})$$

A.8 Approximation of Firm Profit Changes after Exporting

The firm j 's profit function is:

$$\Pi_{jt} = R_{jt} - (P_{jt}^L L_{jt} + P_{jt}^M M_{jt}). \quad (\text{A40})$$

Based on the definition of raw markup $\tilde{\mu}_{jt} \equiv R_{jt}/(P_{jt}^L L_{jt} + P_{jt}^M M_{jt})$, the profit function can be expressed as:

$$\begin{aligned}\Pi_{jt} &= \frac{R_{jt}(\tilde{\mu}_{jt} - 1)}{\tilde{\mu}_{jt}} \\ \pi_{jt} &= r_{jt} + \ln \left(\frac{\tilde{\mu}_{jt} - 1}{\tilde{\mu}_{jt}} \right).\end{aligned}\tag{A41}$$

The approximation of the firm profit changes after exporting is shown as follows:

$$\begin{aligned}\Delta\pi^{exp} &= \pi^{exp=1} - \pi^{exp=0} \\ &= (r^{exp=1} - r^{exp=0}) + \left\{ \left[\ln \left(\frac{\tilde{\mu} - 1}{\tilde{\mu}} \right) \right]^{exp=1} - \left[\ln \left(\frac{\tilde{\mu} - 1}{\tilde{\mu}} \right) \right]^{exp=0} \right\} \\ &= (r^{exp=1} - r^{exp=0}) + \ln \left(\frac{\tilde{\mu}^{exp=1} - 1}{\tilde{\mu}^{exp=0} - 1} \right) - \ln \left(\frac{\tilde{\mu}^{exp=1}}{\tilde{\mu}^{exp=0}} \right) \\ &= \Delta r^{exp} + \ln \left(1 + \frac{\tilde{\mu}^{exp=1} - \tilde{\mu}^{exp=0}}{\tilde{\mu}^{exp=0} - 1} \right) - \ln \left(1 + \frac{\tilde{\mu}^{exp=1} - \tilde{\mu}^{exp=0}}{\tilde{\mu}^{exp=0}} \right) \\ &\simeq \Delta r^{exp} + \frac{\tilde{\mu}^{exp=1} - \tilde{\mu}^{exp=0}}{\tilde{\mu}^{exp=0} - 1} - \frac{\tilde{\mu}^{exp=1} - \tilde{\mu}^{exp=0}}{\tilde{\mu}^{exp=0}} \\ &= \Delta y^{exp} + \Delta p^{exp} + \frac{1}{\tilde{\mu}^{exp=0} (\tilde{\mu}^{exp=0} - 1)} \Delta \tilde{\mu}^{exp} \\ &= \Delta y^{exp} + \Delta p^{exp} + \frac{1}{(\alpha_L + \alpha_M) \tilde{\mu}^{exp=0} (\tilde{\mu}^{exp=0} - 1)} \Delta \mu^{exp}\end{aligned}\tag{A42}$$

A.9 Quantity Survey Data

The output quantity data is from the Annual Product Quantity Survey of the Industrial Enterprises (APQSIE). In this dataset which spans from 2000-2008, the firms have reported their product type (by both product name and 5-digit product code) and the output quantity for each product type with the quantity unit in each year.²⁷ Besides, since this dataset is also collected by the NBS of China, the firms share the same ID and name that can be easily merged to our main ASIE dataset to acquire the information shown in the Table 1 (e.g. total sales, labor employed, material expenditure, capital stock, etc.), and the final merged sub-sample with the quantity information is also ranged from 2000 to 2006.

A.9.1 Quantity Adjustment w.r.t. Units

Before merging the APQSIE data with the ASIE data, we need to process the quantity output data to make sure the output quantity of different firms is as comparable as possible. To generate the true price data, we only keep those firms with a single product. Then we adjust the quantity w.r.t. the quantity units.

²⁷The product code is missed in the years 2006–2008, and the quantity unit is missed in the years 2000–2003, which requires us to make use of the information in other years to complete the missing information in the years above.

Specifically, we only keep the observations with the quantity unit information. First, for each unit in the list (individual, piece, kilowatt, pair, unit, ton, block, set, square meter, handle, branch, volume, strip, slice, cubic meter, meter, door), we multiply the quantity by 10,000 if the quantity unit is “ten thousand units” and then replace the quantity unit with the corresponding unit without “ten thousand”. Next, we multiply the quantity by 1,000 if the quantity unit is “kilometer” and then replace the quantity unit with “meter”. If the quantity unit is “core kilometer”, we multiply the quantity by 1,000 and then replace the quantity unit with “meter”. Similarly, if the quantity unit is “pair kilometer”, we multiply the quantity by 2,000 and then replace the quantity unit with “meter”. We also multiply the quantity by 1,000 if the quantity unit is “ton”, “total ton”, “comprehensive ton”, or “evaporation ton”, and then replace the quantity unit with “kilogram”. If the quantity unit is “pair”, we multiply the quantity by 2 and then replace the quantity unit with “unit”. Finally, we drop observations where the quantity unit is “kilowatt-hour”, which is only used for measure the electricity power generation in the sample. Besides, since we use the sales revenue as the output in the main sample, after merging the APQSIE with the main sample, we first generate the output price by output value divided by the output quantity. Then, we use the sales revenue divided by the output price to derive the quantity.

Moreover, we trim the quantity and price data by the largest and smallest 1% to avoid the influence of some extreme values. Eventually, the dataset includes 118,671 firms with 321,280 observations, among which there are 19,838 firms (16.72%) are exporters.

A.9.2 Input Deflators

Another important adjustment we have made is the deflators. Since we only have the material expenditure data, we borrow the industry-year level input deflator P_{jt}^M from [Brandt et al. \(2017\)](#) to get the deflated (log) material m_{jt} . However, we notice the fact that the (log) quantity output shares a very low correlation with the deflated (log) material, compared to the (log) sales revenue: The difference between y_{jt} and the r_{jt} is the removal of the price

Table A1: Correlations between Outputs & Material

	(1)	(2)
	y_{jt}	r_{jt}
m_{jt}	0.305 (0.0000)	0.977 (0.0000)

Note: P -values in parentheses.

information, which is positively correlated with the material as shown from the Table A1. Inspired by [Li and Zhang \(2022\)](#), we consider the difference is caused by the correlation between input quality and output quality. y_{jt} only contains the product quantity information, where the product quality information is removed, and we are using the industry-year-level material deflator to proxy m_{jt} , where the input quality information is still contained, so r_{jt} shows a higher correlation with m_{jt} . If we directly use the industry-year-level-deflated material in the estimation, the results would be biased. To illustrate this, we write down the

(log) production function with both the output quality and input quality²⁸:

$$y_{jt} + \chi_{jt}^Y = \alpha_L l_{jt} + \alpha_M (m_{jt} + \chi_{jt}^M) + \alpha_K k_{jt} + \omega_{jt} + \xi_{jt}^X, \quad (\text{A43})$$

where y_{jt} is the quantity output, with the output quality χ_{jt}^Y , and m_{jt} is the quantity input, with the input quality χ_{jt}^M . If we estimate the production function using the quantity output information from the APQSIE and material expenditure deflated by the industry-year-level material deflator $\bar{P}_{jt}^M$, we are actually estimating the function:

$$y_{jt} = \alpha_L l_{jt} + \alpha_M \ln \left(\frac{E_{jt}^M}{\bar{P}_{jt}^M} \right) + \alpha_K k_{jt} + \omega_{jt} + (\xi_{jt}^X - \chi_{jt}^Y), \quad (\text{A44})$$

$$= \alpha_L l_{jt} + \alpha_M (m_{jt} + \chi_{jt}^M - \bar{p}_{jt}^M) + \alpha_K k_{jt} + \omega_{jt} + (\xi_{jt}^X - \chi_{jt}^Y), \quad (\text{A45})$$

$$= \alpha_L l_{jt} + \alpha_M \tilde{m}_{jt} + \alpha_K k_{jt} + \omega_{jt} + \tilde{\xi}_{jt}^X, \quad (\text{A46})$$

where $\tilde{m}_{jt}$ is the quality-inclusive material. However, as illustrated in [De Loecker et al. \(2016a\)](#) and [Li and Zhang \(2022\)](#), the input quality is (positively) correlated with the output quality, so the error term $\tilde{\xi}_{jt}^X$ is correlated with the $\tilde{m}_{jt}$, biasing the estimation results. Besides, even though we take the quality problem into account, the firm-level input price, which is partially contained in the $E_{jt}^M / \bar{P}_{jt}^M$, is still positively correlated with the output price, so the endogeneity problem is still severe.

To avoid this endogeneity problem, we make use of the output price to construct a firm-year-level input price index based on the assumption of the linear relationship between input quality and output quality:

$$\tilde{P}_{jt}^M = \bar{P}_{jt}^M \times \frac{P_{jt}}{\bar{P}_{jt}}, \quad (\text{A47})$$

where $\bar{P}_{jt}^M$ and $\bar{P}_{jt}$ are industry-year-level deflators w.r.t. material and output as in [Brandt et al. \(2017\)](#), and P_{jt} is the output price calculated by output value divided by the output quantity described before. After using $\ln(E_{jt}^M / \tilde{P}_{jt}^M)$ to proxy the material, the correlation between the quantity output and material becomes 0.997.

During the estimation, except for controlling for the city, ownership, and industry fixed effect as in the traditional methods, we also control for the product code $\times$ year fixed effect in the first stage to control for the remaining inconsistency between the quantity output of different firms. Furthermore, in order to be consistent with the estimation in the main sample, we construct the new industry-year-level material deflator $\tilde{P}_{jt}^M$ by calculating the corresponding mean of $\tilde{P}_{jt}^M$ and the new industry-year-level output deflator $\tilde{P}_{jt}$ by calculating $\bar{P}_{jt} \times \tilde{P}_{jt}^M / \bar{P}_{jt}$ for both our methods and the traditional methods using the revenue information.

A.10 External Validation: Colombia

In order to further validate the effectiveness of our method and findings (regarding the export

²⁸Detailed discussions can also be found in [De Loecker et al. \(2016a\)](#).

premium and IRS), we analyze the Colombian manufacturing industry using our approach. This Colombian plant-level dataset is collected by the National Administrative Department of Statistics (Departamento Administrativo Nacional de Estadística, DANE). In this dataset, each plant reported multiple accounting indicators, such as total sales, total employment, the value of intermediate consumption, the total book value of fixed assets, and export value, covering the period from 1981 to 1991. However, key quantitative information regarding both output and intermediate inputs is missing.

Additionally, as reported in Table A2, 10% to 16% of plants exported each year. Thus, as discussed in our previous sections, the price decline associated with exporting—documented by Garcia-Marin and Voigtländer (2019) using Colombian data from 2001 to 2013—can bias the estimates of the productivity gains from exporting. To address this, we apply our method to revisit the Colombian data.

Table A2: Plants in the Colombian Sample

Year	Number of Plants	Number of Exporters
1981	6,054	683 (11.28%)
1982	6,279	701 (11.16%)
1983	5,660	632 (11.17%)
1984	5,700	602 (10.56%)
1985	5,838	671 (11.49%)
1986	6,072	733 (12.07%)
1987	6,236	717 (11.50%)
1988	6,369	731 (11.48%)
1989	6,593	796 (12.07%)
1990	6,485	874 (13.48%)
1991	6,199	1,035 (16.70%)
Total	67,503	8,175 (12.11%)

Column (1) of Table A3 reports the estimation results based on the Colombian data. There are three main findings. First, exporting has a positive and statistically significant effect on productivity, as indicated by the parameter β_{exp}^{ω} . Specifically, exporting raises Colombian plants' productivity by about 0.7% on average ($\hat{\beta}_{exp}^{\omega} = 0.007$ in both Columns (1)–(2) of Table A3). This effect is comparable to the 0.8% productivity gain estimated for Chinese exporters. The smaller productivity gain in Colombia may reflect differences in industrial composition or the export strategies of firms in the two countries. For example, the Chinese export sector during the studied period was characterized by rapid industrial upgrading and integration into global value chains, which may have amplified the productivity benefits of exporting. In contrast, Colombian exporters may have faced structural constraints, such as less developed infrastructure or lower levels of technology adoption, which could limit the productivity-enhancing effects of exporting. Despite the smaller effect, the result still highlights the role of exporting in improving firm-level efficiency.

Second, exporting substantially increases a plant’s market size, as captured by the parameter β_{exp}^D . The demand-parameter estimate is $\hat{\beta}_{exp}^D = 0.112$ (SE = 0.0146) under LP and 0.119 (SE = 0.0185) under ACF; after 10% symmetric truncation, the mean firm-level demand effect $\beta_{exp}^D/(\mu - 1)$ is approximately 29.6% (LP) and 22.7% (ACF), substantially larger than the 15.4% observed for Chinese exporters. This suggests that Colombian firms may experience a more pronounced market expansion when entering export markets, potentially due to the smaller size of their domestic market relative to China. Firms in smaller economies may rely more heavily on exports to access larger international markets, leading to a greater relative increase in demand. The larger market size provides Colombian exporters with opportunities to achieve economies of scale and higher profits.

Third, as in China’s case, Colombian plant-level production also exhibits substantial IRS after correcting for firm heterogeneity in markup. The total returns to scale, S , are estimated at 1.239 under LP and 1.362 under ACF, both higher than the 1.103 estimated for Chinese firms. This result suggests that Colombian firms benefit more from scale economies in production. The larger RTS in Colombia may reflect structural differences in production technology or firm size distribution compared to China.

As shown in column (2), these three findings are also robust with the alternative method provided by [Akerberg et al. \(2015\)](#). Both Colombian and Chinese exporters exhibit IRS and experience productivity gains and demand expansion.

As a comparison, columns (3)–(4) report the estimation results using the widely adopted approaches in the literature based on revenue data to estimate the revenue production function without correcting for firm heterogeneity in markup. Specifically, column (3) assumes a constant markup, following [Klette and Griliches \(1996\)](#), while column (4) uses deflated revenue without any markup adjustment.

As shown in the table, without correcting for firm heterogeneity in prices (and markup), the traditional methods find a negligible or even negative effect of export on revenue productivity (−0.3% to 0.2%), which is an order of magnitude lower compared to our results. Both columns (3) and (4) exhibit increasing returns to scale (IRS), and the magnitude of RTS is very close to the estimates reported in [Garcia-Marin and Voigtländer \(2019\)](#) (1.036–1.038), which used quantity output information but abstracted from the input price problem²⁹ to estimate the Colombian data from 2001 to 2013.

However, as [De Loecker et al. \(2016a\)](#) highlights, it is crucial to account for input price variations by using output-input price linkage information. Failing to address this issue can lead to an underestimation of RTS, as discussed in Appendix [A.9](#). Comparing the RTS estimates in columns (1)–(2) ($S = 1.239$ under LP and 1.362 under ACF) with those in columns (3)–(4) ($S = 1.091$ –1.093), it is evident that traditional methods underestimate RTS, and this issue is not limited to China’s case.

²⁹In [Garcia-Marin and Voigtländer \(2019\)](#), plant-product fixed effects are controlled for. However, as documented by [Grieco et al. \(2016a\)](#) and [Li and Zhang \(2022\)](#), quality upgrading in output is often correlated with quality upgrading in intermediate inputs. As a result, firm-level heterogeneity may not be fully accounted for when firms begin exporting.

Table A3: Estimation Results of Colombian Data (1981–1991)

Parameter	(1)	(2)	(3)	(4)
	Ours		K&G(1996)	Deflated
	LP	ACF	LP	LP
<i>Panel A: Estimates of Production Function</i>				
β_{exp}^{ω} (Productivity Effect)	0.007 (0.0029)	0.007 (0.0052)	0.001 (0.0012)	-0.002 (0.0015)
β_{exp}^D (Demand Parameter)	0.112 (0.0146)	0.119 (0.0185)	0.075 (0.0096)	0.054 (0.0088)
RTS	1.239 (0.0137)	1.362 (0.0503)	1.093 (0.0051)	1.091 (0.0064)
RTSV	1.202 (0.0150)	1.333 (0.0488)	1.071 (0.0071)	1.083 (0.0083)
α_L	0.377 (0.0121)	0.422 (0.0576)	0.281 (0.0068)	0.278 (0.0065)
α_M	0.825 (0.0088)	0.911 (0.0166)	0.789 (0.0068)	0.804 (0.0062)
α_K	0.037 (0.0035)	0.028 (0.0045)	0.023 (0.0034)	0.009 (0.0030)
<i>Panel B: The Implied Demand Effect of Export</i>				
Mean	0.296	0.227	0.075	0.054
Median	0.280	0.220		
SD	0.105	0.064		
IQR	0.153	0.095		
IDR	0.281	0.173		
Observations	67,503	67,503	67,503	67,503

Note 1: This table re-estimates the model on plant-level data from the Colombian manufacturing census (1981–1991, Section 7.5), as an external-data validation. Columns (1)–(2) implement our proposed method with the LP and ACF first stages respectively. Because the U.S. NTR tariff gap used as an instrument in the Chinese-data analysis is not available for Colombia, both columns report benchmark specifications without instrumentation.

Note 2: Columns (3)–(4) implement alternative procedures from the literature. Column (3) follows Klette and Griliches (1996), imposing a constant industry-level markup. Column (4) deflates revenue with an industry-level price index without any markup adjustment.

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across plants with differing market power. For Columns (1)–(2), we report the mean, median, SD, IQR, and IDR of this plant-level effect after symmetric 10% truncation at both tails. For Columns (3)–(4), the demand parameter enters the revenue production function as a constant elasticity, so Panel B reports its point estimate in the mean row only.

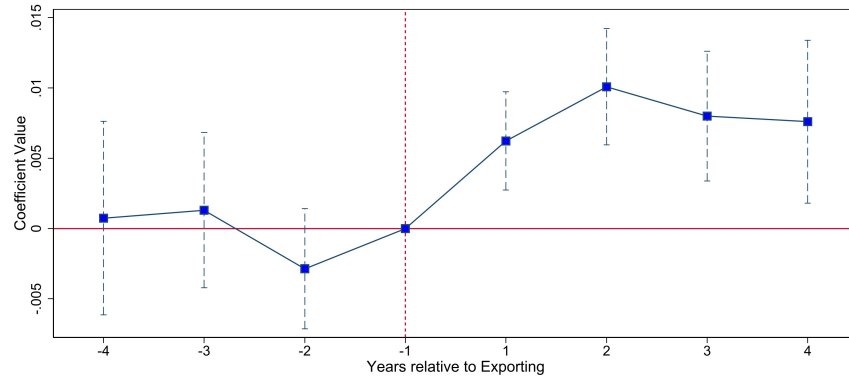
Note 4: Standard errors clustered at the plant level are reported in parentheses.

The gains-from-export decomposition for the Colombian sample is reported and discussed in Section 7.5 of the main text; see Table 10 there. In brief, exporting reduces the marginal cost of Colombian plants by 5.894%, of which 89.481% (5.274 pp) operates through increasing returns to scale and 10.519% (0.620 pp) through the productivity-premium channel. The markup rises by 2%, consumer prices fall by 3.894%, and exporter profits rise by 35.103%, all

in the same direction as the Chinese sample but quantitatively larger. The IRS channel is therefore even more dominant in the Colombian economy.

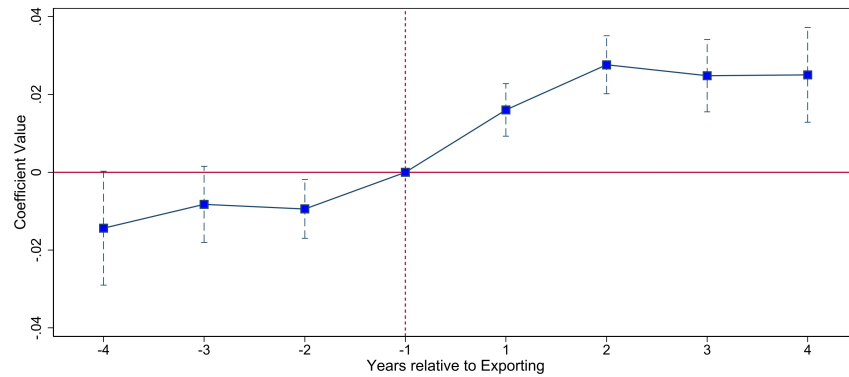
B Appendix - Figures and Tables

Figure B1: Dynamic Exporting Effect on Markup (ACF)



Note: The range represents 95% confidence interval of the parameter estimates, and the standard error is clustered at the firm level.

Figure B2: Dynamic Exporting Effect on Markup for Translog Production Function



Note: The range represents 95% confidence interval of the parameter estimates, and the standard error is clustered at the industry level to address RTS correlations within the industry. Firm size and capital intensity are controlled in the estimation.

Table B1: Implied changes in TFPR

(1)	(2)	(3)	(4)
Markup	Productivity	IRS	TFPR
1%	0.080%	-1.134%	-0.054%

Table B2: Exporting Effect on Markup (ACF)

	(1)	(2)
	Markup	Markup
D_{jt}^{exp}	0.008*** (0.0015)	0.010*** (0.0015)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.326	0.326

Note: Standard errors (clustered at the firm level) in parentheses.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B3: Gains from Export Decomposition (ACF)

Marginal Cost ↓ 1.711%				
through IRS	through Productivity	Markup	Consumer Price	Exporter Profits
↓ 1.098%	↓ 0.613%	↑ 1%	↓ 0.711%	↑ 25.346%

Note 1: This table reports the gains-from-export decomposition under ACF first-stage estimation, analogous to Table 6 for the LP first stage. According to equation (22), exporters' marginal cost declines by 1.711% on average within the first year of exporting. Of this decline, 64.173% (1.098 percentage points) operates through increasing returns to scale (IRS), and the remaining 35.827% (0.613 percentage points) is attributable to the export productivity premium.

Note 2: As reported in Table B7, the markup rises by 1% on average upon exporting. Combining the markup increase with the marginal-cost decline through the decomposition in equation (23) implies that the consumer price falls by 0.711% on average.

Note 3: Applying the decomposition in equation (26), exporter profits increase by 25.346% on average.

Note 4: Compared with the LP-based decomposition in Table 6, the ACF estimates deliver a slightly smaller total marginal-cost decline (1.711% versus 1.854%) and a correspondingly smaller consumer-price decline (0.711% versus 0.854%), but the qualitative pattern is identical: the IRS channel dominates the productivity channel, the markup rises by the same 1%, and exporter profits gain by essentially the same magnitude (25.346% versus 25.384%). The conclusions are therefore not sensitive to the choice of first-stage productivity estimator.

Table B4: Estimation Results for Translog Production Function

Parameter	(1) (IV)	(2) (Benchmark)
Panel A: Estimates of Production Function		
β_{exp}^ω (Productivity Effect)	0.018 (0.0035)	0.007 (0.0024)
β_{exp}^D (Demand Parameter)	0.088 (0.0101)	0.107 (0.0027)
RTS	1.690 (0.0584)	1.663 (0.0174)
RTSV	1.584 (0.0705)	1.552 (0.0276)
α_L	0.084 (0.0436)	0.075 (0.0014)
α_M	0.817 (0.0173)	0.803 (0.0007)
α_K	0.044 (0.0350)	0.038 (0.0008)
α_{LL}	0.023 (0.0069)	0.020 (0.0015)
α_{MM}	0.034 (0.0047)	0.036 (0.0008)
α_{KK}	0.006 (0.0073)	0.010 (0.0007)
α_{LM}	-0.005 (0.0060)	-0.003 (0.0017)
α_{LK}	-0.023 (0.0090)	-0.021 (0.0012)
α_{MK}	0.008 (0.0118)	0.001 (0.0009)
Panel B: The Implied Demand Effect of Export		
Mean	0.111	0.140
Median	0.107	0.135
SD	0.027	0.035
IQR	0.040	0.051
IDR	0.072	0.093
Observations	787,247	1,234,292

Note 1: This table assesses the robustness of our main results to relaxing the Cobb–Douglas functional-form assumption by adopting a translog production function. Columns (1)–(2) implement the procedure developed in Section 3.3; the IV and benchmark specifications follow Table 3.

Note 2: Panel A reports the first-order coefficients ($\alpha_L, \alpha_M, \alpha_K$) and second-order coefficients ($\alpha_{LL}, \alpha_{MM}, \alpha_{KK}, \alpha_{LM}, \alpha_{LK}, \alpha_{MK}$) of the translog production function. Because output elasticities are firm-specific under translog, S and S^V are reported as the sample means of firm-level returns to scale.

Note 3: Panel B reports the implied export-induced demand effect $\beta_{exp}^D/(\mu - 1)$ from the demand system in equation (6), which is heterogeneous across firms with differing market power. The panel reports the mean, median, SD, IQR, and IDR of this firm-level effect after symmetric 10% truncation at both tails.

Note 4: Standard errors clustered at the firm level are reported in parentheses.

Table B5: Exporting Effect on Markup for Translog-form Production Function

	(1)	(2)
	Markup	Markup
D_{jt}^{exp}	0.032*** (0.0021)	0.026*** (0.0021)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.377	0.378

Note: Standard errors (clustered at the industry level) in parentheses.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B6: Exporting Effect on Log-Markup (LP)

	(1)	(2)
	Log-Markup	Log-Markup
D_{jt}^{exp}	0.009*** (0.0012)	0.010*** (0.0012)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.345	0.346

Note 1: Standard errors (clustered at the firm level) in parentheses.

Note 2: Control variables are in log-form.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B7: Exporting Effect on Log-Markup (ACF)

	(1)	(2)
	Log-Markup	Log-Markup
D_{jt}^{exp}	0.009*** (0.0012)	0.010*** (0.0012)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.345	0.346

Note 1: Standard errors (clustered at the firm level) in parentheses.

Note 2: Control variables are in log-form.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B8: Exporting Effect on Log-Revenue

	(1)	(2)
	Log-Revenue	Log-Revenue
D_{jt}^{exp}	0.211*** (0.0045)	0.210*** (0.0045)
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	1,234,292	1,234,292
Adjusted R^2	0.823	0.823

Note 1: Standard errors (clustered at the firm level) in parentheses.

Note 2: Control variables are in log-form.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B9: Exporting Effect on Log-Markup (Colombian Data)

	(1)	(2)
	Log-Markup	Log-Markup
D_{jt}^{exp}	0.023*** (0.0049)	0.024*** (0.0049)
Firm Size (L)		YES
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	67,503	67,503
Adjusted R^2	0.489	0.496

Note 1: Standard errors (clustered at the firm level) in parentheses.

Note 2: Control variables are in log-form.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table B10: Exporting Effect on Log-Revenue (Colombian Data)

	(1)	(2)
	Log-Revenue	Log-Revenue
D_{jt}^{exp}	0.283*** (0.0179)	0.275*** (0.0179)
Capital Intensity		YES
Firm FE	YES	YES
Year FE	YES	YES
Observations	67,503	67,503
Adjusted R^2	0.951	0.951

Note 1: Standard errors (clustered at the firm level) in parentheses.

Note 2: Control variables are in log-form.

* $p < .10$, ** $p < .05$, *** $p < .01$